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Dynamic Capabilities and Organizational Performance: A Meta-Analytic Evaluation and Extension

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ABSTRACT We move the dynamic capabilities view (DCV) forward in two important ways by meta-analysing prior empirical studies. First, we evaluate the two core theoretical tenets of the DCV: (1) Dynamic capabilities are positively related to performance, and (2) this relationship is stronger in industries with higher levels of technological dynamism. We find support for the former ($r_c = 0.296$) but not for the latter, though results suggest the existence of moderators. Second, we theorize and demonstrate empirically that higher-order dynamic capabilities are more strongly related to performance than lower-order dynamic capabilities, lower-order dynamic capabilities partially mediate the relationship between higher-order dynamic capabilities and performance, and dynamic capabilities contribute more to performance in developing economies than in developed economies. These findings illustrate how the nature of the dynamic capability and the economic context in which it is utilized shape its value, thus offering a more nuanced conceptualization of the dynamic capabilities-performance relationship.

Keywords: dynamic capabilities, evolutionary economics, meta-analysis, organizational performance, resource-based theory

INTRODUCTION

Although the dynamic capabilities view has emerged as a flagship perspective within strategic management, the dispersed nature of this research stream has inhibited its paradigmatic and coherent development (Vogel and Güttel, 2013). This state of the literature has led Barreto (2010, p. 277) to call for 'a move toward more *selection*- and

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retention-oriented stages, that is, with a consolidation of the main construct and a capitalization on previous research in a more structured, focused way'. Indeed, we have seen numerous attempts in the form of literature reviews and bibliographic analyses to bring together the proliferating theoretical and empirical works on the dynamic capabilities-performance relationship (e.g., Ambrosini and Bowman, 2009; Di Stefano et al., 2010; Helfat and Martin, 2015; Vogel and Güttel, 2013; Wang and Ahmed, 2007).

As the field evolves towards maturity, theoretical work is converging around two main tenets of the dynamic capabilities view (Helfat et al., 2007; Helfat and Peteraf, 2014; Peteraf et al., 2013): (1) dynamic capabilities contribute to organizational performance, and (2) the value of dynamic capabilities is more pronounced in environments characterized by rapid technological change (Salvato and Rerup, 2011; Teece, 2014; Teece et al., 1997). Yet, as empirical evidence for these assertions has been equivocal (e.g., Schilke, 2014a; Wang et al., 2014; Wilden and Gudergan, 2014; Wilden et al., 2013; Winter, 2003), it is appropriate at this stage of the field's evolution to take stock of 'what we know', extend and refine the dynamic capabilities view in ways that advance our understanding of the divergent results, and propel a focused agenda for future research. Towards that end, in this paper we meta-analysed 82 samples encompassing 22,320 organizations.[1]

Meta-analysis is a rigourous statistical technique that accounts for statistical artifacts while aggregating empirical findings to 'discern whether relationships exist and provide estimates of their size' (Crook et al. 2008, p. 1142). It is well suited to advance theory within the dynamic capabilities view because any theory will 'stand or fall not on the basis of whether its key constructs can be verified, but upon whether its predictions correspond to reality observed for populations of firms' (Godfrey and Hill, 1995, p. 530). In that way, as Crook *et al.* (2008, p. 1152) explain in their meta-analysis of the resource-based view, '[t]he major advantages of meta-analysis are that it offers a much more accurate estimate of the level of support for a theory than other methods for assessing a research stream, and that it can test theory that is difficult to assess through other means'.

Meta-analysis is also useful in extending and refining existing theory because it can 'identify previously unknown factors that help explain the wide variety of findings that often arise in research streams centered on broad constructs' (Crook et al., 2008, p. 1143). Importantly, since the dynamic capabilities view is inherently about unique types of organizational capabilities and the utility they provide in different contexts, theoretical advances in our understanding of the heterogeneous performance outcomes of dynamic capabilities can develop by further looking into these key elements – the nature of the capability and the role of the environment in which the firm operates (Zahra et al., 2006). Accordingly, we utilize the resource-based and evolutionary logics underpinning the dynamic capabilities view to advance our understanding of the dynamic capabilities–performance relationship in two theoretically relevant ways.

First, just as there are different classes of resources (Crook et al., 2008), there are different levels of dynamic capabilities. A particularly important yet underdeveloped aspect of the dynamic capabilities view is the hierarchical ordering of capabilities (Ambrosini and Bowman, 2009; Collis, 1994). In fact, the origins of the dynamic capabilities view highlight the unique nature of dynamic capabilities – these are capabilities that change capabilities (Winter, 2003). The performance implications of dynamic capabilities at different levels, however, remain understudied. We conceptually and empirically differentiate between higher-order and lower-order dynamic capabilities, and argue that the former generates higher performance benefits, directly and indirectly via lower-order dynamic capabilities. As such, we explicate how the hierarchical ordering of dynamic capabilities matters to organizational performance.

Second, our study offers clarity regarding the role of the external environment in the dynamic capabilities-performance link by not only including technological dynamism as a contextual moderating factor, but also considering the role of economic context in shaping the value of dynamic capabilities. To date, the dynamic capabilities literature has been preoccupied with the role of technological dynamism in the task environment, largely ignoring other dimensions of the environment. This is an important gap, as the value of any organizational capability 'must be understood in the specific market context within which a firm is operating' (Barney, 2001, p. 52). By focusing too narrowly on industry effects, the impact of the broader context might be missed. We draw on international business research integrating economic institutions with resource-based and evolutionary logics (e.g., Dunning and Lundan, 2010; Peng, 2003; Peng et al., 2009) to argue and demonstrate that dynamic capabilities are more valuable in developing economies compared to developed economies.

In sum, this study contributes to the dynamic capabilities literature in three broad areas. First, we consolidate empirical studies to *evaluate* the two core theoretical tenets of the dynamic capabilities view, namely that (1) dynamic capabilities contribute to organizational performance, and (2) the value of dynamic capabilities is enhanced in technologically dynamic industries. Doing so also allows us to rigourously evaluate *whether, how much*, and *when* dynamic capabilities contribute to performance. We find support for the first tenet ($r_c = 0.30$) but not for the second, though substantial variability in the main effect size suggests the existence of moderators. Second, we contribute to the literature by specifying how the hierarchical ordering of dynamic capabilities and the economic context serve as contingencies that alter their value. We find that higher order dynamic capabilities are more strongly related to performance than lower-order dynamic capabilities, lower-order dynamic capabilities partially mediate the relationship between higher-order dynamic capabilities and performance, and dynamic capabilities contribute more strongly to performance in developing economies as compared to developed economies. These extensions offer a more nuanced conceptualization of the relationship between dynamic capabilities and performance. Third, we test our moderation hypotheses by employing a novel meta-analytic regression technique, controlling for the impact of methodological factors. By focusing on performance implications and contingencies, as well as the role of research design, we are able to move the field toward clarifying the outcomes of dynamic capabilities, and facilitate the accumulation of consistent evidence in support of a robust theory of dynamic capabilities.

THEORY AND HYPOTHESES

Dynamic Capabilities: Resource-based and Evolutionary Underpinnings

Capabilities are 'a firm's capacity to deploy *resources*, usually in combination, using organizational processes, to effect a desired end' [italics in original] (Amit and

Schoemaker, 1993, p. 35). Winter (2003) emphasized the role of collectively-learned, routine-based activity as a defining feature of capabilities. Thus, *ad-hoc* problem solving or disjointed entrepreneurial improvisations are not capabilities, unless they initiate the emergence of some pattern over time (Moliterno and Wiersema, 2007; Schreyögg and Kliesch-Eberl, 2007; Winter, 2012). Helfat and Winter (2011, p. 1244) succinctly summarize the various definitions of organizational capability, noting that a capability is in place when 'the organization (or its constituent parts) has the capacity [i.e., is able] to perform a particular activity in a reliable and at least minimally satisfactory manner'.

Scholars have distinguished *dynamic* capabilities from *ordinary* capabilities. Ordinary/operational/zero-order capabilities allow an organization to make a living in the present, while dynamic capabilities alter the way an organization makes its living (Helfat and Winter, 2011). Dynamic capabilities allow the firm to alter the resource base, change ordinary capabilities, and/or initiate change in the organization's external environment (Arend and Bromiley, 2009; Barrales-Molina et al., 2014; Helfat and Winter, 2011; Teece, 2007). For instance, Zahra et al. (2006, p. 921) noted that 'new routine for product development is a new substantive capability but the ability to *change* such capabilities is a dynamic capability'. Helfat et al. (2007) build on seminal works in the field (e.g., Eisenhardt and Martin, 2000; Teece et al., 1997) and define dynamic capabilities as the 'capacity of an organization to purposefully create, extend, or modify its resource base' (p. 4).[2]

Thus, the purposeful alteration of *resource* configurations is characteristic of resource-based theory (Barney, 1991; Wernerfelt, 1984), while the attention to change-oriented routines and learning reveals the evolutionary roots of dynamic capabilities (Helfat and Peteraf, 2009; Teece, 2014; Teece et al., 1997; Zollo and Winter, 2002). Hence, the contribution of dynamic capabilities to competitive advantage can be evaluated using both resource-based theory's value-rarity-inimitability-substitutability framework and evolutionary economics whereby an advantage may arise according to the extent to which a dynamic capability enables continuous learning, adaptation, and thus increased alignment with the environment (Nelson and Winter, 1982; Winter, 2012; Zott, 2003). Indeed, as Augier and Teece (2009, p. 418) noted, dynamic capabilities are in part about introducing valuable and unique novelty into the resource base, and in part about promoting and shaping continuous morphing (Rindova and Kotha, 2001). By combining resource-based and evolutionary logics, the dynamic capability view focuses on capabilities that may confer a competitive advantage by adding unique value to the firm through systematic *change*, particularly in industries characterized by rapid technological change (Peteraf et al., 2013; Teece et al., 1997).

Core Theoretical Tenets of the Dynamic Capabilities View

Tenet I: dynamic capabilities contribute to performance. Barreto (2010) argues that dynamic capabilities systematically solve problems, thus allowing the organization 'to make timely and market-oriented decisions' (p. 271), and to introduce innovative changes to the resource base (Ambrosini and Bowman, 2009; Schilke 2014b; Teece, 2014; Wang and Ahmed, 2007). Dynamic capabilities constantly improve ordinary capabilities (i.e., problem-solving for the present), which is more likely to result in superior efficiency

(Zollo and Winter, 2002). At the same time, dynamic capabilities involve building new resources or problem-solving capabilities for the future (Danneels, 2015), which may confer a competitive advantage by initiating change in the competitive environment (Sarkar et al., 2001; Teece, 2014).

Some scholars have argued that dynamic capabilities may not necessarily create the right resource configurations (Ambrosini and Bowman, 2009; Eisenhardt and Martin, 2000) and entail cost (Lavie, 2006; Pablo et al., 2007). However, others have pointed out that they tend to be valuable (Peteraf et al., 2013), and, because dynamic capabilities are 'idiosyncratic in their details' (Eisenhardt and Martin, 2000, p. 1105) and learned as organizations respond to their environment (Winter, 2012), there could be an appreciable and difficult-to-imitate value added to the most experienced firms (Peteraf et al., 2013).[3] Zott (2003) argues that systematic change to the resource base may result in significant performance differentials because these activities allow the organization to accumulate knowledge about *how to change and with fewer costs*, and thus increase congruence with the environment.

Supporting both resource-based and evolutionary benefits of dynamic capabilities, empirical evidence for the positive performance implications of dynamic capabilities abound (e.g., Morgan et al., 2009; Schilke, 2014b). For instance, Fang and Zou (2009) found that dynamic marketing capabilities of international joint ventures lead to superior performance because such capabilities allow these ventures to 'track changes in [and respond to] the consumer environment' (p. 749). Similarly, Heimeriks et al. (2012) demonstrate the positive impact of risk management and tacit knowledge transfer practices, as dynamic capabilities, on post-acquisition financial performance. Furthermore, Stadler et al. (2013) provide evidence for a positive impact of dynamic capability attributes on successful development of valuable and hard-to-imitate resources in the upstream oil industry. In sum, based on prior resource-based and evolutionary theorizing and evidence, we expect that dynamic capabilities should, on the whole, be positively related to organizational performance.

Hypothesis 1: Dynamic capabilities are positively related to organizational performance.

Tenet II: the enhancing role of industry technological dynamism. As rapid external technological changes erode the value of existing capabilities and productive resources (Collis, 1994; Tripsas, 1997; Wang and Ang, 2004), increasing technological dynamism would require changes in the organization's resource base to ensure its alignment with the environment. While organizations in any environment may possess and benefit from dynamic capabilities (Marcus and Anderson, 2006; Protogerou et al., 2014; Schilke, 2014a), the advantages would be more pronounced in technologically dynamic industries because the continuous generation of new resource configurations makes it possible for the organization to achieve congruence with new technological conditions (Drnevich and Kriauciunas, 2011; Teece, 2014; Zahra and George, 2002). Weerawardena et al. (2007, p. 294) maintain that dynamic capabilities allow the firm to 'develop cutting-edge knowledge intensive products, paving the way for their accelerated market entry'.

Thus, in the face of frequent technological change, dynamic capabilities should have more value because such a context increases the opportunity to exercise dynamic capabilities (Schilke, 2014a) and thus better cover the costs of developing them (Zahra et al., 2006). Also, from an evolutionary perspective, the continuous and purposeful morphing that dynamic capabilities facilitate (Rindova and Kotha, 2001) encourages learning and diminishes inertial forces, both of which have been identified as critical success factors in technologically turbulent industries (Ambrosini and Bowman, 2009). Therefore, we expect the following:

> *Hypothesis 2*: Technological dynamism moderates the positive relationship between dynamic capabilities and performance such that the relationship between dynamic capabilities and organizational performance will be stronger among firms operating in task environments characterized by higher technological dynamism.

Theoretical Extensions to the Dynamic Capabilities View

As explicated above, the dynamic capabilities view supports an overall positive relationship between dynamic capabilities and performance, though it also recognizes that contingencies may alter the value of dynamic capabilities. We build on this notion to explicate additional moderating factors that may generate substantive heterogeneity in firm performance. We do so by addressing two relevant yet under researched areas within the dynamic capabilities view. First, while the treatment of capabilities within resource-based theory and evolutionary economics envisages them as hierarchically ordered (e.g., Collis, 1994; Danneels, 2002, 2008; Winter, 2003), it is still unclear how the order of dynamic capabilities affects performance, and whether higher-order dynamic capabilities provide performance benefits beyond their alteration of lower-order capabilities (Schilke, 2014b).

Second, while the strategy literature has long recognized the multi-dimensional and multilevel nature of organizational environments (Dess and Beard, 1984; McCarthy et al., 2010), the dynamic capabilities view has largely focused on the technological dimension of the task environment; however, more recent studies have started examining other aspects of the organization's environment (Eisenhardt et al., 2010; Wilden and Gudergan, 2014). We heed Romme et al.'s (2010) call and complement these studies by explicating the role of the economic context in shaping the value of dynamic capabilities. This is particularly relevant to the value of dynamic capabilities because the economic context will determine the nature of competition and change within a country, both of which may shift the return from investment in change-oriented capacities (Hoskisson et al., 2000). As Dunning and Lundan (2010, p. 1227) noted, the economic context 'defines the main instruments whereby firms... encounter uncertainties in the environment', and should thus be 'explicitly incorporated into any analysis of dynamic capabilities'.

Extension I: the hierarchical ordering of dynamic capabilities. Capabilities exist at different levels; the higher the level, the more abstract and complex they become (Easterby-Smith et al., 2009; Schilke, 2014b; Zollo and Winter, 2002). Schilke (2014b, p. 369) explains that 'while learning routines have always been considered an important component of first-order dynamic capabilities..., second-order dynamic capabilities can be thought of as "learning-to-learn" capabilities'. Winter (2003) defines lower–order dynamic capabilities as those effecting change in the resource base or ordinary capabilities (e.g., changes to the production process; Ambrosini et al., 2009), and higher-order dynamic capabilities as those resulting from organizational learning which creates or modifies lower-order dynamic capabilities.

Accordingly, higher-order dynamic capabilities comprise more metaphysical strategic insights and complex capacities that are more difficult to observe and decode (Collis, 1994). From a resource-based theory perspective, these features render higher-order dynamic capabilities valuable and harder to imitate (Dyer and Hatch, 2006; Wu et al., 2007). Because increased complexity (and hence inimitability) helps maintain performance differentials, we expect to observe that investment in higher-order dynamic capabilities is likely to be more beneficial compared to investment in lower-order dynamic capabilities (Crook et al., 2008; Krasnikov and Jayachandran, 2008; Lockett et al., 2009).

Further, in an evolutionary sense, while lower-order dynamic capabilities are underpinned by processes of adaptive learning (i.e., changing the resource base in a manner consistent with organizational norms and architecture), higher-order dynamic capabilities are underpinned by fungible processes of generative learning and integration of novel insights, which results in more valuable resource bases for competitive advantage (Argyris, 1977; Jarratt, 2004; Verona, 1999). Higher-order dynamic capabilities 'correspond to performing more general activities that govern or integrate the lower-level activities' (Maritan, 2001, p. 527). For instance, companies may engage in R&D to find new and better ways to do R&D. Therefore, higher-order dynamic capabilities offer performance benefits in the form of generating and integrating valuable lower-order dynamic capabilities and resources, which enhances evolutionary fit (Roberts and Grover, 2012; Teece, 2007, 2014; Winter, 2003).

Moreover, higher-order dynamic capabilities also generate value by facilitating more effective ad hoc problem solving (Bingham and Eisenhardt, 2011; Schilke, 2014b) and path-breaking change in the environment (Teece, 2014). Their higher order nature entails change in the organization as a whole in an attempt to harmonize with the environment. Indeed, Ambrosini et al. (2009) and Zahra et al. (2006) explain that higher-order dynamic capabilities are more transformational – they can change the way in which the firm solves its problems. In that sense, higher-order dynamic capabilities contribute to organizational performance directly and via the generation of lower-order dynamic capabilities (Hult and Ketchen, 2001). Taken together, we expect the following:

Hypothesis 3a: The nature of the dynamic capability moderates its effect on organizational performance such that higher-order dynamic capabilities have a stronger positive relationship with performance than lower-order dynamic capabilities.

Hypothesis 3b: Lower-order dynamic capabilities partially mediate the relationship between higher-order dynamic capabilities and organizational performance such that higher-order dynamic capabilities affect performance directly and also indirectly through lower-order dynamic capabilities.

Extension II: the role of economic context. The international business literature maintains that dynamic capabilities that prove to be valuable in one economic context may be less valuable and plentiful in other contexts (Peng et al., 2009). This is because the set of economic institutions governing a business system will determine the costs and benefits of investment in particular resources and capabilities (Drnevich and Kriauciunas, 2011; Schilke, 2014a). Consequently, the economic context should moderate the value derived from dynamic capabilities. In particular, we expect that the performance benefits conferred by dynamic capabilities would be more pronounced in developing economies for two main reasons. First, the relatively lower development of market-based institutions in developing economies renders dynamic capabilities rarer and more valuable compared to in developed economies. Second, the changing economic landscape in developing economies enhances the value of capacities to adapt to changing economic conditions, especially since many firms lack the willingness to enact such changes, making the contribution of dynamic capabilities to performance more pronounced.

Augier and Teece (2009), as well as Dunning and Lundan (2010, p. 1226), explicate that if the new routines and innovative changes provided by dynamic capabilities become widely diffused, 'they no longer confer a unique advantage to the originating firm'. The implication is that economic contexts in which diffusion is less efficient may enhance the advantage dynamic capabilities may entail. A more developed economic context is likely to increase the transparency of the sources of firm capabilities (Jacobides and Winter, 2005). This is because 'where market-based transactions dominate, firms are likely to gain easier access to the innovations that have allowed other firms to gain lower transaction costs or to provide higher value' (Dunning and Lundan, 2010, p. 1238). Since the 'rules of the game' are well defined (He and Cui, 2012) and organizations compete largely based on market-based competencies, information asymmetries among market participants are relatively low, competitive advantage is more transient, and gains from improvements to the resource base can be small (McGrath, 2013). Hence, dynamic capabilities are more widespread and thus generate less marginal value.

Conversely, in developing economies, integrated hierarchical solutions, such as business groups, are dominant (Guillen, 2000; Khanna and Palepu, 1997), so innovative firms have less incentive to increase the transparency of competitive practices. Consequently, other firms in the economy find it more difficult to identify and imitate organizational innovations. In addition, Dixon et al. (2010) argue that the economic system in many developing economies had discouraged innovation and experimentation, such that dynamic capabilities are less common. 'Not only are the historical resources of firms inefficient (Uhlenbruck et al., 2003), but also the inherited norms, values, and assumptions underlying economic activity are completely different to those of a market economy' (Dixon et al., 2010, p. 420). Such institutional inefficiencies limiting vigorous market-based competition are more likely to inhibit firms from investing in dynamic capabilities (Chari and Banalieva, 2015; Shinkle and Kriauciunas, 2012). Even after privatization, many organizations 'continue in their old ways' (Dixon et al., 2010, p. 423) according to older institutional templates, and core rigidities are more common.[4]

Taken together, these prior studies indicate that dynamic capabilities are likely to be more plentiful in developed economies due to the more efficient diffusion of best practices and the relative lack of willingness among developing-economy firms to develop dynamic capabilities and purposefully change in innovative ways (Morrison et al., 2008; Yiu et al., 2007). Therefore, while dynamic capabilities are important in both contexts, their marginal value is different. Although dynamic capabilities allow firms to keep with competition in developed economies, they are less likely to give them a competitive edge. In developing economies, however, the marginal benefit of dynamic capabilities is likely to be higher due to their rarity and relatively slower diffusion. Consequently, greater performance benefits would accrue to firms that do invest in dynamic capabilities (Chari and David, 2012; Malik, 2008). As Fang and Zou (2009) explain, the development of dynamic capabilities is particularly more relevant to explaining competitive advantage or superior firm performance in developing economies.

Furthermore, in many developing economies, we see transition towards market-based arrangements, which brings about changes to industry structure, competitive dynamics, and market demand (Kim et al., 2010; Luo, 2003a,b; Schilke, 2014a). According to Peng et al. (2007, p. 206), 'while firms in developed economies do experience some environmental dynamism… the scale and scope of such dynamism pale in comparison with the comprehensive changes of the "rules of the game" experienced by firms in China'. Again, dynamic capabilities are valuable in both contexts, but the intensity and frequency of economic change in developing economies provides further opportunities to exercise dynamic capabilities and thus renders them a stronger driver of superior performance (Fang and Zou, 2009; Schilke, 2014a). This is particularly the case since many competitive advantages in emerging economies are based on network relationships (including membership in business groups) and government ties (Hoskisson et al., 2000). As the economic context changes, there are necessary changes in firms' asset structure and orientation (Guillen, 2000) as firms need to tap into new demand curves (Fang and Zou, 2009; Slater and Narver, 1994) while their operational capabilities are eroded. For instance, the personalized nature of transactions in developing economies requires a different set of capabilities to achieve a competitive advantage, compared to that required in increasingly market-based economic contexts (Peng, 2003).

Consequently, organizations that invest in dynamic capabilities are not only more likely to sense the need for change, but also better positioned to leverage such changes into innovation-based advantages (Chari and David, 2012). Weerawardena et al. (2007) explain that dynamic capabilities are useful especially when markets emerge. Dynamic capabilities can facilitate the change required in such a context, as they entail learning and innovative change to the resource base that translate into better fit with the environment. As Dixon et al. (2010) note, dynamic capabilities are particularly important for firms facing significant economic changes, an uncertain institutional environment, and poorly developed markets (Uhlenbruck et al., 2003). Since many 'stay in their old ways' in the face of economic change, those that invest in change-oriented capabilities should be in a better position to achieve a greater competitive advantage and thus better performance (Fang and Zou, 2009). These arguments lead us to propose the following:

Hypothesis 4: The economic context moderates the relationship between dynamic capabilities and performance, such that the relationship is weaker (stronger) in developed (developing) economies.

DATA AND METHODOLOGY

Sample

The sample selection procedure comprised several stages. First, using the key terms 'dynamic capabilities'/'dynamic capability' and 'performance'/'competitive advantage', we searched for articles published in peer reviewed journals as of December, 2013 but later than 1997 – the publication year of the seminal piece by Teece *et al.* (1997) – in the ABI/INFORM and EconLit databases. We also conducted a series of manual searches in management, international business, marketing, and entrepreneurship journals, and examined reference sections of published review articles (e.g., Barreto, 2010; Easterby-Smith et al., 2009; Wang and Ahmed, 2007).[5]

Next, we excluded articles that did not include methodological keywords, such as data, empirical, test, statistical, finding(s), result(s), or evidence in their abstract (Newbert, 2007), as well as those that did not have the words Teece, Eisenhardt, *or* Helfat in the text. Filtering results based on such content and methodological keywords is a useful and efficient procedure to hone in on the most relevant and empirical articles for a quantitative review, especially when a generic search yields a very high number of results – in our case, over four thousand (see David and Han, 2004; Newbert, 2007). We also made an effort to remove entries where we identified both the relationship and data examined across two studies were the same.

Then, we examined whether articles included required statistical information (e.g., correlations, sample size) about the relationship between dynamic capabilities and competitive advantage or performance. Finally, to reduce potential reverse causality, studies had to measure dynamic capabilities and performance at least simultaneously; studies that examined the effect of performance on dynamic capabilities (e.g., Moliterno and Wiersema, 2007) were not included in our sample. These procedures yielded a final set of 82 samples (in 79 studies) encompassing 22,320 organizations. The articles included in the meta-analytic calculations are marked with an asterisk in the References section.

Coding

Coding was conducted to accompany procedures outlined in previous meta-analyses in the strategy literature (e.g., Crook et al., 2008; Kirca et al., 2011; Palich et al., 2000). To code for Hypothesis 1, we obtained three statistics from each article: sample size, correlation between dynamic capabilities and performance, and reliability levels for dynamic capabilities and performance. However, not all studies reported reliability statistics, so for constructs from such studies we used the average reliability from studies that did report reliability. Further, some studies may include several dynamic capabilities and several performance measures. For the purposes of this test, we averaged these effects, such that each sample is represented once (Hunter and Schmidt, 2004).

With regards to Hypothesis 2, technological dynamism is often reflected in 'an increase in the rate of technological change and its diffusion throughout that industry' (Simerly and Li, 2000, p. 38). Prior research indicates that industries where R&D spending and proportion of knowledge workers are higher tend to be more turbulent (Thornhill, 2006; Zahra and Neubaum, 1998). As noted by Bierly and Daly (2007, p. 499), '[i]n high-technology industries, which have been referred to as hypercompetitive and high-velocity industries, the ability to frequently challenge the status quo with new, breakthrough ideas is critical for firm success'. In high-tech industries, process and product technologies shift not only more frequently but also in higher magnitudes as compared to other industries (Wang et al., 2014). Following Daily et al. (2005) and Wang et al. (2014), we coded samples of firms as operating in high-technology industries when the primary industry was aerospace, computer hardware, computer software, biotechnology, or telecommunications. We refer to this dichotomous variable as *technological dynamism* (1 – high tech industry, 0 – otherwise).[6]

To code for Hypothesis 3a, we drew from Collis (1994, p. 145) and distinguish between dynamic capabilities that introduce 'dynamic improvement to the activities of the firm' (i.e., lower-order) and 'more metaphysical strategic insights' (i.e., higher-order; Schilke 2014b). Higher-order dynamic capabilities are characterized by higher complexity and generative change as compared to lower-order dynamic capabilities. We actively assessed each dynamic capability in our sample studies; dynamic capabilities that met these two criteria were considered higher-order (1 – *higher-order dynamic capability* and 0 – otherwise). Three members from the research team participated in coding the dynamic capabilities in our sample studies. The coders are researchers familiar with the dynamic capabilities literature and were debriefed about the criteria discussed above. In cases where there were discrepancies between coders, the discrepancy was resolved through discussion.

We focused on how constructs were measured, not labelled. For instance, Hsu and Wang (2012) and Cui and Jiao (2011) both use 'dynamic capability' as their primary construct. Whereas the former measures the construct with increases in R&D and marketing spending, the latter utilizes items about the firm's reconfiguration capacity (Cui and Jiao, 2011, p. 391) which better captures a generative capacity to alter the resource base. Similarly, the latter is characterized by higher complexity because it measures a capacity requiring a large number of interdependencies within the organization (Reed and DeFillippi, 1990). Conversely, a dynamic capability measured as some variant of R&D expenditure likely does not represent well a complex and generative capacity across firms (Crook et al., 2008). In assessing the generative nature of higher-order dynamic capabilities, we looked for measures that capture changes to the way the organization refines the resource base. As an example, Schilke (2014a) uses 'new product development' as one of the primary constructs for dynamic capabilities and measures the construct with items such as 'we extend product range'. In that case, the utilized measure captures adaptive activities in the product development area but does not imply more overarching, generative change.

To test Hypothesis 3b, we created a three-cell matrix composed of meta-analytic estimates of the intercorrelations between higher-order dynamic capabilities, lower-order dynamic capabilities, and performance. The correlation between higher-order and lower-order dynamic capabilities (r = 0.49) was calculated ad hoc from six studies in our sample which included both types. The correlation between higher-order dynamic capabilities and performance (r = 0.34), as well as the one between lower-order dynamic capabilities and performance (r = 0.21), were calculated based on our sample.

To test Hypothesis 4, we coded whether the sample came from a developed economy (1) or otherwise (0) (Kirca et al., 2012). In our case, such a distinction was not only theoretically but also empirically appropriate because the vast majority of samples from developing economies were from emerging or transition economies (e.g., India, China, Chile). In such economies, changes in economic conditions are particularly frequent and substantial, while many firms find it hard to let go of imprinted practices stemming from previous ideologies (Marquis and Raynard, 2015; Shinkle and Kriauciunas, 2012). Similar to previous studies (e.g., Cuervo-Cazurra, 2006; Di Giovanni, 2005; Lawler et al., 2011), we used the CIA World Factbook list of developed countries. This source defines developed countries as high-income, market-oriented economies with mainly democratic regimes, most of which are OECD members. This definition captures well the essence of our argumentation for Hypothesis 4.[7]

Table I presents the substantive moderator classifications for each study.

To test our moderation hypotheses, we coded several control variables that were expected to affect the dynamic capabilities-performance relationship. This was also useful in assessing the effect of methodological factors on effect size heterogeneity (Certo et al., 2006). First, recent studies are more likely to employ established measures as compared to studies in the nascent phase of the dynamic capabilities literature. Therefore, to examine whether the relationship between dynamic capabilities and performance has been changing over time as the research stream matures, we split the sample studies into two groups: before (coded as 0) and after 2007 (coded as 1) (Palich et al., 2000).[8] We label this variable *recent publication*.

Second, we focused on the source and independence of data, and the nature of performance and dynamic capability measures. Support may be weaker when studies use archival performance or dynamic capabilities measures as well as when independent dynamic capabilities and performance data sources are utilized. While single-source perceptual data may cause potential upward bias (Williams et al., 2010), it is at least equally plausible that this is due to the lower reliability of archival measures, given that these are often only crude proxies whereas perceptual measures tend to be more fine-grained (Venkatraman and Ramanujam, 1986). Therefore, we coded the following variables: *dynamic capability perceptual* (1 – measures based on survey data, 0 – archival), *performance perceptual* (1 – measures based on survey data, 0 – archival), and *data source dependence* (1 – data for dynamic capabilities and performance obtained from a single respondent, 0 – otherwise).

Some performance variables may be more susceptible to appropriation by various stakeholders (Coff, 1999), which may reduce the observed effect size (Crook et al., 2008). Thus, we expected dynamic capabilities to be more strongly related to *non-appropriable performance* outcomes (e.g., market share, competitive advantage, sales growth) as compared to more appropriable measures (e.g., ROA). We coded this variable into two main categories (1 – non-appropriable, 0 – appropriable) based on measures employed in each study. For some studies, several performance indicators were reported; all were

Table I. Moderator classification by study

Study	*Dynamic capability*	*Higher order dynamic Capability*	*Technological dynamism*	*Developed economy*
Arend, 2013	• Dynamic Capabilities • Ethic Focused Dynamic Capabilities	• Yes • No	Mixed	Yes
Arthurs and Busenitz, 2006	• Assessment of Weaknesses and Threats • Opportunity Identification • Reconfiguration	• Yes • Yes • Yes	High	Yes
Bingham et al., 2007	• Heuristics	• Yes	Mixed	Yes
Blesa and Ripollés, 2008	• Marketing Capabilities	• No	Mixed	Yes
Blome et al., 2013	• Supply Chain Agility	• Yes	Low	Yes
Bock et al., 2012	• Creative Culture	• Yes	Mixed	Mixed
Caloghirou et al., 2004	• Dynamic Capabilities	• Yes	Low	Yes
Capron and Mitchell, 2009	• R&D Capabilities	• No	High	Mixed
Chakrabarty and Wang, 2012	• R&D Intensity • Internationalization	• No • No	Mixed	Mixed
Chen and Chang, 2013	• Green Dynamic Capabilities	• Yes	High	No
Chen and Lien, 2013	• Technological Opportunism	• Yes	Mixed	No
Chen et al., 2012	• Dynamic Capabilities	• Yes	High	Yes
Cheng and Chen, 2013	• Dynamic Innovation Capabilities	• Yes	High	No
Chien and Tsai, 2012	• Dynamic Capabilities	• Yes	Low	No
Cui and Jiao, 2011	• Dynamic Capabilities	• Yes	Mixed	No
Danneels, 2012	• Marketing Competence • R&D Competence	• No • No	Low	Yes
Deeds et al., 2000	• Research Team Capabilities • R&D Management Capabilities	• No • No	High	Yes
Drnevich and Kriauciunas, 2011	• Dynamic Capabilities	• Yes	Mixed	No
Eggers, 2012	• Firm Breadth • Breadth of New Funds • Firms New Funds In Same Category	• No • No • No	Low	N/A
Eng and Spickett-Jones, 2009	• Marketing Capabilities	• No	Low	No
Ettlie and Pavlou, 2006	• Interfirm New Product Development Partnership Dynamic Capabilities	• No	Low	Yes
Fang and Zou, 2009	• Dynamic Marketing Capabilities	• Yes	Low	No
Fernández-Mesa et al., 2013	• Organizational Learning Capability • Design Management Capability - Basic Skills • Design Management Capability - Specialized Skills • Design Management Capability - Involving Others • Design Management Capability - Organizational Change	• Yes • No • No • Yes • Yes • Yes	Low	Yes

Table I. *Continued*

Study	*Dynamic capability*	*Higher order dynamic Capability*	*Technological dynamism*	*Developed economy*
	• Design Management Capability - Innovation Skills			
Filatotchev and Piesse, 2009	• R&D Intensity • Export Intensity	• No • No	Mixed	Yes
Franco et al., 2009	• Market Pioneering • Market Responding • Technological Capabilities	• No • No • No	High	N/A
García-Morales et al., 2012	• Organizational Learning • Organizational Innovation	• Yes • No	Low	Yes
Heimeriks et al., 2012	• Risk Management Practices • Tacit Knowledge Transfer Practices	• No • No	Mixed	Yes
Hsu and Sabherwal, 2012	• Dynamic Capabilities	• Yes	Low	No
Hsu and Wang, 2012	• Dynamic Capability	• No	High	No
Huang et al., 2012	• Process/Path/Position	• No	High	No
Huh et al., 2008	• Organizational Capabilities	• No	Mixed	Yes
Hung et al., 2010	• Organizational Dynamic Capability	• Yes	High	No
Jantunen et al., 2005	• Reconfiguring Capabilities	• No	Low	Yes
Jiang et al., 2010	• Firm Diversity	• No	Low	Mixed
Kale and Singh, 2007	• Alliance Learning Process • Alliance Function	• Yes • No	High	Yes
Karim, 2009	• Degree Of Reorganization	• No	High	Mixed
Khalid and Larimo, 2012	• Alliance Management Capability • Marketing Planning And Implementation Capability • Alliance Learning Capability	• No • No • Yes	High	Mixed
Khatri et al. (2014) - unpublished	• Human Resource Management Capability • Information Technology Capability	• No • No	Low	No
Kim et al., 2013	• Advanced Manufacturing Technology • E-Procurement • Market Flexibility	• No • No • No	Mixed	N/A
Kor and Mahoney, 2005	• R&D Deployments • Marketing Deployments	• No • No	High	Yes
Kotha et al., 2011	• Branching of Technology Capabilities	• No	High	Yes
Lee et al., 2010	• Dynamic Resource Reconfiguration Capability	• Yes	High	Yes
Lee et al., 2011	• Marketing Program Implementation	• No	Low	Yes
Li and Liu, 2014	• Dynamic Capability	• Yes	Mixed	No
Liu and Hsu, 2011	• Capability Upgrade and Exploitation	• No	High	No
Lu and Ramamurthy, 2010	• IT Innovation	• No	Mixed	Yes
Luo, 2003	• Local Responsiveness • Control flexibility	• No • No	Low	No

Table I. *Continued*

Study	*Dynamic capability*	*Higher order dynamic Capability*	*Technological dynamism*	*Developed economy*
Malik and Kotabe, 2009	• Organizational Learning • Reverse Engineering • Manufacturing Flexibility	• Yes • Yes • Yes	Low	No
Menguc and Auh, 2006	• Market Orientation	• Yes	Low	Yes
Morgan et al., 2009	• Marketing Capabilities	• No	Mixed	Yes
Morgan et al., 2012	• Architectural Marketing Capabilities • Specialized Marketing Capabilities	• No • Yes	Mixed	Yes
O'Cass and Sok, 2012	• Marketing Capability • Innovation Capability	• No • No	Low	No
Pavlou and El Sawy, 2010	• Dynamic Capability	• Yes	High	Yes
Protogerou et al., 2012	• Dynamic Capability	• Yes	Low	Yes
Rice et al., 2015	• Organizational Processes	• No	Low	Yes
Roberts and Grover, 2012	• Customer Agility	• Yes	High	Yes
Rothaermel and Hess, 2007	• Innovative Output	• No	High	Mixed
Salge and Vera, 2013	• Incremental Learning	• No	Low	Yes
Santos-Vijande et al., 2011	• Organizational Learning	• Yes	Low	Yes
Santos-Vijande et al., 2013	• Brand Management System	• No	High	Yes
Schilke, 2014a	• Alliance Management • New Product Development	• No • No	Low	Yes
Shamsie et al., 2009	• Differentiated Replication • Differentiated Renewal	• No • No	Low	Yes
Singh et al., 2013	• Technological Capability • R&D Capability • Innovation Capability • Alliance Capability • Human Resource Capability	• No • No • No • No • No	Low	No
Song et al., 2005	• Dynamic Marketing/Technological Capabilities	• Yes	Mixed	Yes
Stadler et al., 2013	• Dynamic Capability	• Yes	High	Yes
Uhlenbruck, 2004	• Parent MNE Acquisition and Regional Experience	• No	Mixed	No
Uhlaner et al., 2013	• External Sourcing Capability • Employee Involvement in Renewal Activities	• No • No	Mixed	Yes
Vickery et al., 2013	• Supply Chain Integration	• No	Low	Yes
Webb and Schlemmer, 2008	• Dynamic Capability	• Yes	High	Yes
Wei and Wang, 2010	• Sensing • Seizing • Supply Chain Visibility	• Yes • Yes • Yes	Mixed	No

Table I. *Continued*

Study	*Dynamic capability*	*Higher order dynamic Capability*	*Technological dynamism*	*Developed economy*
Wilden et al., 2013	• Dynamic Capabilities	• Yes	Mixed	Yes
Wu et al., 2007	• Dynamic Capabilities	• No	High	No
Wu, 2006	• Dynamic Capabilities	• Yes	High	No
Wu, 2007	• Dynamic Capabilities	• Yes	High	No
Wu, 2010	• Dynamic Capabilities	• Yes	High	No
Yalcinkaya et al., 2007	• Dynamic Capabilities	• Yes	Low	Yes
Yiu and Lau, 2008	• Resource Capital Reconfiguration	• Yes	Low	No
Zhan and Chen, 2013	• Exploitation capability	• Yes	Mixed	No
	• Exploration Capability	• Yes		
Zhan and Luo, 2008	• Capability Exploitation	• Yes	Mixed	No
	• Capability Upgrading	• Yes		

Notes: The designation 'mixed' means that the study provided description of the focal characteristic, but it did not clearly fall into one category within that characteristic. 'N/A' means that the study did not provide sufficient information to determine the classification of the study based on the focal characteristic. Due to space considerations, we do not present the classification for our methodological moderators (i.e., control variables). These details are available from the authors.

obtained and coded separately. Finally, because capabilities are aimed at a particular output, the correlation between dynamic capabilities and performance measures may depend on how proximal the two are (Ray et al., 2004). We therefore actively assessed whether the performance variable is a generic outcome variable (e.g., ROA) or an outcome specifically measuring the output of a corresponding capability (e.g., product innovativeness as a function of a dynamic product development capability). We refer to this variable as *proximal outcome* (1 – proximal outcome, 0 – otherwise).

ANALYSIS AND RESULTS

Analytical Procedure

We analysed the data in three steps. First, in testing Hypothesis 1 we followed procedures established by Hunter and Schmidt (2004), and calculated a sample-size-weighted mean estimate of the correlations (r) between dynamic capabilities and performance. Next, we calculated *unreliability-corrected* correlations (r_c), which we use for interpretation (Hunter and Schmidt, 2004). We also calculated the statistical significance (i.e., 95 per cent confidence intervals), and generalizability (i.e., 80 per cent credibility intervals) for the correlation (Whitener, 1990). If a 95 per cent confidence interval does not include zero, then the meta-analysed correlation coefficient is significantly different from zero. If an 80 per cent credibility interval does not include zero, we can infer that the vast majority of the correlations are larger than zero for positive correlations or smaller than zero for negative ones (Whitener, 1990). The standard deviation for the corrected correlation provides a measure of variation in the relationship of interest, and serves as the

Table II. Meta-analytic relationship between dynamic capabilities and performance

Parameter	*Value*
K	82
N	22,320
r	0.253
r_c	0.296
SD	0.224
95% confidence interval	0.25:0.35
80% credibility interval	0.01:0.58
Q-statistic	990.22 ($p < 0.01$)

Notes: K, number of samples; N, number of companies; r, sample-weighted correlation; r_c, sample-weighted correlation, corrected for unreliability; SD, standard deviation of r_c.

basis for the credibility intervals. Credibility intervals are also indicative of the possible existence of moderators; wide credibility intervals or credibility intervals that span zero might indicate that some contingencies distinguish between several groups of correlations (Whitener, 1990).

During that stage, we also assessed the extent to which publication bias may affect the obtained effect sizes in our sample (Harrison et al., 2014). To address potential concerns, we first estimated the fail-safe N in our sample, and determined that it would take approximately 178 unpublished studies with near-zero dynamic capabilities-performance correlations to potentially reduce the effect size to 'small' (i.e., $r_c = 0.10$; see Orwin, 1983 and Rosenberg, 2005). Next, we conducted a trim and fill procedure, consistent with Duval and Tweedie (2000a,b). This procedure first generates a funnel plot with the horizontal axis representing the effect size of each sample, and the vertical axis representing the inversed standard error of each sample. Due to the random distribution of sampling error, an asymmetrical funnel plot indicates the potential existence of publication bias. According to Harrison et al. (2014, pp. 7–9),

> 'The technique then reintroduces the trimmed samples and imputes or fills the missing samples needed to create a symmetrical distribution. The extent of potential publication bias is established through the number of imputed samples necessary to create a symmetrical distribution (ik) and the difference in the observed effect size among published studies and the "trimmed" effect size'.

In our case, the results of that analysis indicated that zero studies were missing on the left hand side of the plot and that ten studies were missing on the right side of the funnel. Filling the sample with potential unpublished studies would increase the main effect size from 0.30 (see Table II) to 0.36. The corrected positive effect size remains significant at the 95 per cent confidence level [0.31, 0.42]. Taken together, these findings suggest that while publication bias does seem to exist in dynamic capabilities research, it is not a major concern.

In the second step, we conducted weighted regression analyses with random effects models in order to test the moderation hypotheses. We chose random effects models for two main reasons. First, in meta-analyses, there are two potential sources of sampling errors: sampling errors within studies and sampling errors between studies. Fixed effects models assume that the population variance across studies is zero, while random effects models account for this variance (Hunter and Schmidt, 2000; Schmidt et al., 2009). This variation in population parameters may take the form of theoretical or methodological differences across studies, which when ignored in a fixed effect model has been shown to yield narrower confidence intervals (Schmidt et al., 2009) and increased Type I error rates (Hunter and Schmidt, 2000). Second, random effects models also account for the possibility that there are correlations between moderators (Borenstein et al., 2010; Geyskens et al., 2009).

For each relationship between dynamic capabilities and performance, we included the moderators in the regression if there were at least three studies for each moderator category. For all meta-regression analyses, we used the package 'metafor' in R (Viechtbauer, 2010). Following Hedges and Olkin (1985), we transformed the correlations into Fisher's z prior to analysis. In each regression model, the correlation is used as the dependent variable while the different moderators and controls serve as independent variables. For each model, two fit statistics are calculated: $Q_{m,}$ which follows a chi-square distribution and indicates whether the overall model is significant, and Q_r, which also follows a chi-square distribution but indicates whether residual heterogeneity is significant (i.e., non-redundant variability remains to be explained).

As a third step, we tested Hypothesis 3b using the ad hoc correlation matrix as input for structural equation modelling with LISREL. We relied on the reliability-uncorrected correlations and allowed only one correlation per sample for each cell in the matrix. Such an approach is more conservative in terms of detecting significant effects. Similarly, in testing this model, we used the harmonic mean of sample sizes as this gives less weight to large samples and provides a more conservative estimate (Viswesvaran and Ones, 1995).

Results

Table II reports the results of the meta-analysis testing Hypothesis 1. Hypothesis 1 received support – the corrected correlation between dynamic capabilities and performance was positive ($r_c = 0.296$) and the 95 per cent confidence interval did not include zero (0.25:0.35). Nevertheless, the wide credibility interval (0.01:0.58), which is based on the rather high standard deviation (0.224), suggests that the focal relationship may be contingent on other factors. Further, the Q-statistic is significant for this model and the percentage of variance in effect-size estimates explained by artifacts (e.g., sampling error variance, unreliability, range restriction) is well below 75 per cent (i.e., 8.39 per cent; Hunter and Schmidt, 1990), both of which indicate that heterogeneity in effects is substantial and likely influenced by moderators. This is not surprising given the construct breadth of the dynamic capabilities research stream.

Results of the moderator regression analyses are presented in Table III. Model 1 includes only the six control variables, and reveals that *data source dependence* and *proximal outcome* are positive and significant, as expected. Model 2 tests Hypothesis 2, while Models 3, 4, and 5 present tests of Hypotheses 3a and 4. As can be seen in Model 2, Hypothesis 2 was not supported as the coefficient of *technological dynamism* was not significant.

Table III. Meta-regression results for moderator analysis

Variable	*Model 1*	*Model 2*	*Model 3*	*Model 4*	*Model 5*
Control variables:					
Recent publication	−0.01 (0.05)	−0.01 (0.04)	0.02 (0.04)	0.03 (0.04)	0.02 (0.05)
Dynamic capability perceptual	0.10 (0.06)	0.04 (0.07)	0.10 (0.06)	0.05 (0.07)	0.01 (0.08)
Performance perceptual	0.02 (0.08)	−0.08 (0.09)	−0.02 (0.08)	−0.01 (0.10)	−0.03 (0.10)
Data source dependence	0.17 (0.06)**	0.30 (0.08)**	0.16 (0.06)**	0.24 (0.09)**	0.29 (0.09)**
Non-appropriable performance	0.06 (0.04)	0.09 (0.05)*	0.06 (0.04)	0.05 (0.04)	0.07 (0.05)
Proximal outcome	0.13 (0.04)**	0.12 (0.04)**	0.11 (0.04)**	0.09 (0.04)*	0.09 (0.05)
Hypothesized moderators:					
Technological dynamism		−0.07 (0.04)			−0.06 (0.06)
Higher-order dynamic capability			0.08 (0.04)*		
Developed economy				−0.14 (0.04)**	−0.09 (0.05)*
Q_M	109.61**	116.64**	116.78**	102.79**	84.27**
Q_R	1024.71**	604.49**	1006.38**	700.04**	414.21**

Notes: Unstandardized estimates are reported. Standard errors reported in parentheses.
Q_R, Q statistic for residual heterogeneity; Q_M, Q statistic for overall moderator model. * $p < .05$, ** $p < .01$.

Model 3 indicates support for Hypothesis 3a as the effect sizes were higher when studies were coded as utilizing *higher-order dynamic capabilities* ($b = 0.08$, $p < 0.05$). Model 4 shows that effect sizes are lower when studies were conducted in a context of a *developed economy* ($b = -0.14$, $p < 0.01$), supporting Hypothesis 4. Model 5 includes both *technological dynamism* and *developed economy*, showing that only the latter is a significant moderator ($b = -0.09$, $p < 0.05$). Further, while the significant Q_m statistics throughout all models that these models fit the data well, the significant Q_r values show that substantial variability in effect sizes remains to be explained.

In testing Hypothesis 3b, we specified two path models: one in which higher-order dynamic capabilities influence performance only indirectly through lower-order capabilities and another in which higher-order dynamic capabilities affected both lower-order dynamic capabilities and organizational performance while lower-order dynamic capabilities affected only organizational performance. The first model did not fit the data well with $\chi^2(1) = 211.35$, $p < .001$; $CFI = 0.78$; and $SRMR = 0.10$. In this model, the path between lower-order dynamic capabilities and organizational performance is 0.21, while the indirect effect of higher-order dynamic capabilities on performance is 0.10.

The second model allowed for a direct path from higher-order capabilities to organizational performance. As can be seen in Figure 1, higher-order dynamic capabilities contribute to organizational performance directly (path coefficient = 0.31, $p < 0.01$), in addition to their indirect effect through lower-order dynamic capabilities (indirect effect = .03, $p < .05$). This provides statistical support for Hypothesis 3b, whereby lower-order dynamic capabilities are affected by higher-order dynamic capabilities, but only partially mediate their effect on organizational performance. In addition, consistent

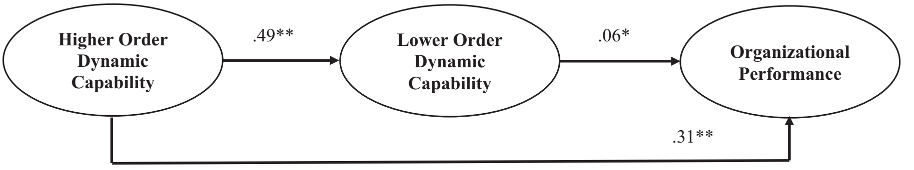

Figure 1. Meta-analytic path model

Notes: Harmonic mean = 2,626; standardized path coefficient are presented; * $p < 0.05$, ** $p < 0.01$.

with Hypothesis 3a, while lower-order capabilities also significantly impact performance (path coefficient = 0.06, $p < 0.05$), the effect of higher-order capabilities is higher when both are allowed direct paths to performance.

We conducted several analyses to assess whether other potential moderators explain observed effect size heterogeneity. We particularly focused on the various types of dynamic capabilities, regardless of hierarchical level, and the various types of performance indicators. We did not include these variables in our main analyses to avoid statistical power and multicollinearity issues. After reviewing the sample studies, we coded dummy variables for the following performance types: *strategic* (i.e., competitive advantage), *financial, operational, innovation*, and *growth*. While such a distinction overlaps with some of our control variables, it is more granular. As for capability types, our review of the sample articles revealed seven types: *generic, technology/R&D, marketing, alliancing, product development, manufacturing*, and *supply chain management*. We therefore coded seven dummy variables in accordance with these types. We found that (a) none of the performance type dummy variables is significant at the 0.05 level; and (b) the dynamic capabilities–performance relationship is stronger when the study utilized new product development capabilities. The other capability types were not significant moderators, which indicates that, for the most part, these variables did not play a significant role in explaining effect size heterogeneity.

DISCUSSION AND CONCLUSIONS

The objective of this study was twofold. First, we aimed to take stock of the dynamic capabilities literature by evaluating its two core assertions: (1) dynamic capabilities have a positive impact on performance, and (b) such impact is stronger in environments characterized by rapid technological change. Meta-analytic results support an overall positive contribution of dynamic capabilities to performance. However, we found that technological dynamism did not moderate the main relationship. Second, we examined two previously uncovered contingencies that may have influenced past studies. We drew on resource-based and evolutionary logic to extend the dynamic capability view by examining the hierarchical nature of dynamic capabilities and the role of the economic context in altering the value of dynamic capabilities. Meta-analytic results support our hypotheses.

Implications for the Dynamic Capabilities View

Our study enriches the dynamic capability view in three important ways. First, we provide an empirical assessment of the two core theoretical tenets within the dynamic capabilities view. Our meta-analysis shows that dynamic capabilities have a positive impact on organizational performance. This finding implies that once sampling and methodological artifacts are accounted for, the core assertion regarding dynamic capabilities' positive performance outcomes is supported by the consolidated empirical evidence. Additionally, our study also provides empirical evidence that counters assertions that the central argument of the dynamic capability view is tautological (Arend and Bromiley, 2009). Results show a moderate effect size (see generally Bosco et al., 2015) between dynamic capabilities and performance ($r_c = 0.30$), supporting the notion that the two are indeed related but distinct constructs.

We draw on resource-based and evolutionary logic to clarify the theoretical mechanisms linking dynamic capabilities to performance. More specifically, dynamic capabilities generate new, valuable, rare and hard-to-imitate resource configurations; by systematically engaging in such change, the organization is also more likely to achieve alignment with the environment. Here, two underlying assumptions are that managers generally initiate change for the better and that the cumulative benefits of dynamic capabilities outweigh their costs (Arend and Bromiley, 2009). This is not necessarily true in all cases; managers do not possess a crystal ball. Yet, what the results tell us based on extant empirical dynamic capabilities literature is that indeed as a general rule, systematic change of the resource base tends to payoff (Zott, 2003). This finding is consistent with some of the recent arguments in the literature (e.g., Helfat and Peteraf, 2014) and heeds Barreto's (2010, p. 274) call to consolidate knowledge of 'the most important relationship in this field'.

The second tenet, the moderating role of technological dynamism, has been much discussed in the literature (Peteraf et al., 2013). While, conceptually, dynamic capabilities are often argued to be more valuable in environments of rapid technological change, some empirical evidence suggests this may not be the case (e.g., Schilke, 2014a). Our meta-analysis allowed us to evaluate this theoretical assertion and the empirical anomalies across a large sample of studies and organizations. Results suggest that technological dynamism is not a significant moderator, which lends support to a thread of research within the dynamic capabilities literature that has expressed doubts about the enhanced value of dynamic capabilities in technologically dynamic industries. Several studies suggest that firms in technologically-dynamic industries possess stronger dynamic capabilities to begin with (Aragon-Correa and Sharma, 2003; Barrales-Molina et al., 2013); therefore, dynamic capabilities in such contexts may not be a differentiating factor. In technologically dynamic industries, where Schumpeterian competition often dominates, developing dynamic capabilities is key to staying in the race (Athreye, 2005) but not necessarily what gives firms a competitive edge (Eisenhardt and Martin, 2000). That is, the competitive advantage provided by dynamic capabilities to firms operating in high-tech industries is perhaps negligible from the outset.

According to Sarkar et al. (2001), in high-tech industries, the costs associated with systematically pioneering change may lead to a first-mover disadvantage or neutralize first mover advantage, even though dynamic capabilities facilitate the creation of value. So, as suggested by Eisenhardt and Martin (2000) and Zollo and Winter (2002), in technologically dynamic industries, there are ample opportunities to exercise dynamic capabilities but their outcomes are unpredictable; in less technologically dynamic industries, dynamic capabilities entail more predictable outcomes but less opportunities to exercise, while having such capabilities in place surely entails costs (also see Rahmandad, 2012; Winter, 2003). This possibly leads to parity in performance outcomes, hence the lack of support for the second tenet in our meta-analysis.

The second way in which our study enriches the dynamic capabilities view is by offering a more nuanced conceptualization of the relationship between dynamic capabilities and performance. We do so in two ways. First, we distinguish between higher-order and lower-order dynamic capabilities. Dynamic capabilities is a broad construct, so it is not surprising that prior theorizing spans a variety of capabilities. However, *explicit* treatments of the hierarchical ordering of *dynamic* capabilities have remained largely absent from the dynamic capability view since Collis' (1994) conceptual study (see Schilke, 2014b and Winter, 2003 for notable exceptions). Our meta-analysis allowed us to spotlight this theoretically-grounded contingency across an array of prior empirical research. Evidently, the hierarchical ordering of dynamic capabilities is a relevant nuance that informs the dynamic capabilities view.

In resource-based terms, higher-order dynamic capabilities generate value both directly and indirectly by enhancing lower-order dynamic capabilities. As for the former, higher-order dynamic capabilities are more fungible, should lead to improved problem-solving within organizations, and are more likely to allow organizations to initiate changes in their environment. This is consistent with Winter (2003, p. 994), who maintained that 'those who invest in routinizing the response to familiar types of change may find themselves disadvantaged relative to more flexible players who have invested in higher-order capabilities'. As for the latter, the capacity to change the way the organization goes about changing its operational capabilities (i.e., higher-order dynamic capabilities) is valuable because it facilitates the learning of novel, unique, and more path-breaking ways to change the way the organization makes its living (see Vergne and Durand, 2011).

These findings present opportunities for further refinement of dynamic capabilities theory. For instance, an examination of additional potential mediating mechanisms (i.e., intermediate outcomes) between higher-order dynamic capabilities and performance would likely enhance our understanding of the consequences of dynamic capabilities. Additionally, while higher-order dynamic capabilities positively affect lower-order dynamic capabilities (and performance), the effect sizes in our path model are not deterministic. Winter (2003) suggests that the benefits of investing in higher-order dynamic capabilities cannot be uniformly or inevitably advantageous. Indeed, Schilke (2014b) shows that alliance learning (a dynamic capability he labels higher-order) positively affects alliance management (a dynamic capability he labels lower-order), but also that the two interact to negatively affect organizational performance. Hence, now that we have presented evidence that the distinction between the two types is relevant to organizational performance, an examination of when the two are complementary or substitutable is warranted. This also implies that future work may consider moderated–mediation models whereby dynamic capabilities affect a sequence of proximal and distal outcomes (Ray et al., 2004), while environmental conditions intervene at various steps to alter the value of dynamic capabilities.

Additionally, we add nuance to the dynamic capabilities-performance link by theoretically specifying and empirically demonstrating the role of the economic context in altering the value of dynamic capabilities. This provides several theoretical insights to dynamic capability theory. For starters, most theorizing to date has conceptualized the environment as unidimensional, with technological change as the focal construct. Yet, several recent contributions have found heterogeneous effects of other dimensions of the environment (e.g., Wang et al., 2014; Wilden and Gudergan, 2014). Our discussion of the economic context as an environmental contingency within a resource-based and evolutionary framework heeds calls by Dunning and Lundan (2010) and Romme et al. (2010) to provide novelty and nuance to the role of the environment within the dynamic capability view.

We find that dynamic capabilities are more strongly related to performance in developing economies. In our view, this result is also underpinned by resource-based and evolutionary logic. The relatively weaker levels of market-based competition and the reluctance of many firms to change their old ways (Shinkle and Kriauciunas, 2012) entail higher value added for firms that systematically seek to improve their resource base in such economies. In market-based settings, improvements yielded by dynamic capabilities may be relatively more transient. Hence, it appears that the performance benefits provided by dynamic capabilities are more apparent in economic environments where they are less prevalent and thus relatively rare. We caution against strong conclusions here as dynamic capabilities may not always change the resource base in the right direction (Ambrosini and Bowman, 2009) and be less effective in the face of too much flux in the economic environment. Nonetheless, these insights provide fertile grounds for dynamic capabilities researchers to uncover additional dimensions of the environment that may shape the value derived from dynamic capabilities, as well as identify boundary conditions for the enhancing role of economic change.

Finally, we make an important methodological contribution to the literature. Using a novel meta-analytic regression technique, which accommodates methodological control variables, we found that research design characteristics play an important and systematic role in differentiating effect sizes. Particularly, as expected, effect sizes were higher when both dynamic capability and performance data came from the same source, and when the performance outcome was proximal to the dynamic capability's intended output (see Ray et al., 2004). While this may not come as a surprise, it is a key finding because such design elements eventually affect inferences made about *theory*. For dynamic capabilities research to make even more coherent progress, methodological rigour is a must. For instance, following the clear terminology laid out in Helfat and Winter (2011) to establish that a capability exists, and that it is in fact dynamic, is a good first step that can be followed by an explanation of how measured attributes capture the focal dynamic capability (e.g., Stadler et al., 2013).

Limitations, Future Research, and Conclusion

Our study is not without limitations. First, the majority of the studies included in our sample were cross-sectional; hence, we must be cautious in making assertions of causality. We encourage scholars to conduct more longitudinal research to addresses this limitation as well as examine the durability of competitive advantage that dynamic capabilities generate. Additionally, we note that the literature has not examined a variety of outcomes that dynamic capabilities may generate. For instance, how do dynamic capabilities affect the firm's ability to restrain rivals, or maximize utility from timing commitment (Makadok, 2011)?

Second, we find that the effect of higher-order dynamic capabilities on performance is partially mediated by lower-order dynamic capabilities. Thus, more studies looking into potential mediating mechanisms in the dynamic capabilities-performance relationship are needed. Similarly, as evident in the credibility interval (0.01:0.58) and significant Q statistics throughout our analyses, substantial variability in terms of moderators remains to be explained. While data limitations inherent in a meta-analysis often prevent the examination of many other potential moderating factors, it also provides a likely fruitful avenue for future research. For instance, Nair et al. (2014) and Makkonen et al. (2014) explore the effect of dynamic capabilities on performance during the 2008 economic crisis. Invoking such additional moderators is a fruitful future research avenue.

Third, while we follow rigorous coding and analytic procedures, we acknowledge the potential limitations entailed in coding prior studies for a meta-analysis, such as the dichotomous and qualitative assessment of dynamic capability hierarchy. Nevertheless, our results indicate that the notion of capability hierarchy is important to the dynamic capabilities view and thus presents a fruitful path forward. Finally, while we followed established guidelines in structuring our sample, it is by no means exhaustive. Our intention was to take stock of existing research and extend the dynamic capability view by drawing on its resource-based and evolutionary foundations. In doing so, we hope to propel more focused theory building and discussion about the performance implications of dynamic capabilities.

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NOTES

[1] A meta-analysis by Krasnikov and Jayachandran (2008) found an overall positive relationship between firm capabilities and performance. Their study, however, did not specifically examine dynamic capabilities and their influence on performance. We discuss the differences between the two types later in this paper. Further, Krasnikov and Jayachandran's (2008) sample includes studies up to 2007. Given the surge of studies in the field of dynamic capabilities in recent years, a more focused and updated meta-analysis is warranted.

[2] Other definitions of dynamic capabilities can be found in the literature (e.g., Baretto, 2010). We focus on Helfat et al.'s (2007) definition as it integrates prior seminal articles (e.g., Eisenhardt and Martin, 2000; Teece et al. 1997). Their work facilitated theoretical convergence, and we use their definition in order to facilitate a manageable discussion.

[3] Since dynamic capabilities in any form are, by definition capabilities, the advantages they may offer can be evaluated using resource-based theory (Peteraf et al. 2013). According to Crook et al. (2008, p. 1144), valuable and rare capabilities provide *at least* a temporary advantage, but because dynamic capabilities are learned cumulatively over time, those imitating them will be at a cost disadvantage (Barney, 1991). Such capabilities can confer sustained competitive advantage. The notion of competitive advantage, in turn, is very closely related to performance (Crook et al., 2008) because the former is usually used to

describe 'relative performance of rivals' in a given competitive environment (Peteraf and Barney, 2003, p. 313).

[4] We are not arguing that firms in developing economies do not possess valuable capabilities, dynamic or otherwise. Literature on developing market multinational corporations (DMNCs) (e.g., Cuervo-Cazurra, 2012; Ramamurti, 2012) documents well the resources and capabilities these firms possess. We argue that developing economy firms are less likely, on average, to develop dynamic capabilities and that these are less likely to be efficiently diffused in the economy. Indeed, many DMNCs have been shown to internationalize particularly in order to obtain ownership advantages and upgrade capabilities at home to catch up to advanced-economy competitors (Chen and Chen, 1998; Dunning, 1980, 2000; Lee and Slater, 2007; Luo and Tung, 2007; Mathews, 2006; Yiu et al., 2007;). For instance, Samsung, a well-known Korean electronics giant, in the past invested heavily in the establishment of joint ventures and production plants in various countries, especially the U.S.A, not only trying not to lag behind, but also to leap to the forefront of DRAM technology (Lee and Slater, 2007). We thank an anonymous reviewer for pointing this out.

[5] We searched in tables of contents of *Academy of Management Journal, Administrative Science Quarterly, British Journal of Management, Entrepreneurship Theory and Practice, Industrial and Corporate Change, International Business Review, Journal of Business Venturing, Journal of International Business Studies, Journal of Management Studies, Journal of Management, Journal of Marketing, Journal of World Business, Management Science, Organization Science, Strategic Entrepreneurship Journal*, and *Strategic Management Journal*. We also tried to obtain unpublished studies in three ways: first, we distributed a call for papers via the Academy of Management (AoM) listservs; second, we posted a call for unpublished papers on the LinkedIn page of AoM; third, we directly emailed authors of relevant papers presented at the Academy of Management and Strategic Management Society annual meetings in the last four years, and asked for a copy of the manuscript, given that it has not been published yet. We were able to obtain one unpublished study.

[6] Several samples included a mix of high-tech and non-high-tech industries (see Table I). These were not included in testing Hypothesis 2 regarding technological dynamism in order to reduce potential measurement error of this construct.

[7] According to the CIA World Factbook (2015), the developed countries list also largely overlaps with the International Monetary Fund's list of advanced economies.

[8] We chose the year 2007 because, with the publication of Teece (2007) and Helfat et al. (2007), a subsequent surge of studies in the field had vastly different conceptualizations on which to draw.

REFERENCES

Ambrosini, V. and Bowman, C. (2009). 'What are dynamic capabilities and are they a useful construct in strategic management?'. *International Journal of Management Reviews*, **11**, 29–49.

Amit, R. and Schoemaker, P. J. (1993). 'Strategic assets and organizational rent'. *Strategic Management Journal*, **14**, 33–46.

Aragon-Correa, J. A. and Sharma, S. (2003). 'A contingent resource-based view of proactive corporate environmental strategy'. *Academy of Management Review*, **28**, 71–88.

Arend, R. J. (2013). 'Ethics-focused dynamic capabilities: a small business perspective'. *Small Business Economics*, **41**, 1–24.*

Arend, R. and Bromiley, P. (2009). 'Assessing the dynamic capabilities view: spare change, everyone?'. *Strategic Organization*, **7**, 75–90.

Argyris, C. (1977). 'Organizational learning and management information systems'. *Accounting, Organizations and Society*, **2**, 113–23.

Arthurs, J. D. and Busenitz, L. W. (2006). 'Dynamic capabilities and venture performance: the effects of venture capitalists'. *Journal of Business Venturing*, **21**, 195–215.

Athreye, S. S. (2005). 'The Indian software industry and its evolving service capability'. *Industrial and Corporate Change*, **14**, 393–418.

Augier, M. and Teece, D. J. (2009). 'Dynamic capabilities and the role of managers in business strategy and economic performance'. *Organization Science*, **20**, 410–21.

Barney, J. (1991). 'Firm resources and sustained competitive advantage'. *Journal of Management*, **17**, 99–120.

Barney, J. B. (2001). 'Resource-based theories of competitive advantage: a ten-year retrospective on the resource-based view'. *Journal of Management*, **27**, 643–50.

Barrales-Molina, V., Bustinza, Ó. F. and Gutiérrez-Gutiérrez, L. J. (2013). 'Explaining the causes and effects of dynamic capabilities generation: a multiple-indicator multiple-cause modelling approach'. *British Journal of Management*, **24**, 571–91.

Barrales-Molina, V., Martínez-López, F. J. and Gázquez-Abad, J. C. (2014). 'Dynamic marketing capabilities: toward an integrative framework'. *International Journal of Management Reviews*, **16**, 397–416.

Barreto, I. (2010). 'Dynamic capabilities: a review of past research and an agenda for the future'. *Journal of Management*, **36**, 256–80.

Bierly, P. E. and Daly, P. S. (2007). 'Alternative knowledge strategies, competitive environment, and organizational performance in small manufacturing firms'. *Entrepreneurship Theory and Practice*, **31**, 493–516.

Bingham, C. B. and Eisenhardt, K. M. (2011). 'Rational heuristics: the 'simple rules' that strategists learn from process experience'. *Strategic Management Journal*, **32**, 1437–64.

Bingham, C. B., and Furr, N. R. (2007). 'What makes a process a capability? Heuristics, strategy, and effective capture of opportunities'. *Strategic Entrepreneurship Journal*, **1**, 27–47.

Blesa, A. and Ripollés, M. (2008). 'The influence of marketing capabilities on economic international performance'. *International Marketing Review*, **25**, 651–73.

Blome, C., Schoenherr, T. and Rexhausen, D. (2013). 'Antecedents and enablers of supply chain agility and its effect on performance: a dynamic capabilities perspective'. *International Journal of Production Research*, **51**, 1295–318.

Bock, A. J., Opsahl, T., George, G. and Gann, D. M. (2012). 'The effects of culture and structure on strategic flexibility during business model innovation'. *Journal of Management Studies*, **49**, 279–305.

Borenstein, M., Hedges, L. V., Higgins, J. and Rothstein, H. R. (2010). 'A basic introduction to fixed-effect and random-effects models for meta-analysis'. *Research Synthesis Methods*, **1**, 97–111.

Bosco, F. A., Aguinis, H., Singh, K., Field, J. G. and Pierce, C. A. (2015). 'Correlational effect size benchmarks'. *Journal of Applied Psychology*, **100**, 431–49.

Caloghirou, Y., Protogerou, A., Spanos, Y. and Papagiannakis, L. (2004). 'Industry-versus firm-specific effects on performance: contrasting SMEs and large-sized firms'. *European Management Journal*, **22**, 231–43.

Capron, L. and Mitchell, W. (2009). 'Selection capability: how capability gaps and internal social frictions affect internal and external strategic renewal'. *Organization Science*, **20**, 294–312.

Certo, S., Lester, R., Dalton, C. and Dalton, D. (2006). 'Top management teams, strategy and financial performance: a meta-analytic examination'. *Journal of Management Studies*, **43**, 813–23.

Chakrabarty, S. and Wang, L. (2012). 'The long-term sustenance of sustainability practices in MNCs: a dynamic capabilities perspective of the role of R&D and internationalization'. *Journal of Business Ethics*, **110**, 205–17.

Chari, M. D. and Banalieva, E. R. (2015). 'How do pro-market reforms impact firm profitability? The case of India under reform'. *Journal of World Business*, **50**, 357–67.

Chari, M. D. and David, P. (2012). 'Sustaining superior performance in an emerging economy: an empirical test in the Indian context'. *Strategic Management Journal*, **33**, 217–29.

Chen, H. and Chen, T. J. (1998). 'Network linkages and location choice in foreign direct investment'. *Journal of International Business Studies*, **29**, 445–67.

Chen, C. W. and Lien, N. H. (2013). 'Technological opportunism and firm performance: moderating contexts'. *Journal of Business Research*, **66**, 2218–25.

Chen, P. L., Williams, C. and Agarwal, R. (2012). 'Growing pains: pre-entry experience and the challenge of transition to incumbency'. *Strategic Management Journal*, **33**, 252–76.

Chen, Y. S. and Chang, C. H. (2013). 'The determinants of green product development performance: green dynamic capabilities, green transformational leadership, and green creativity'. *Journal of Business Ethics*, **116**, 107–19.

Cheng, C. C. and Chen, J. S. (2013). 'Breakthrough innovation: the roles of dynamic innovation capabilities and open innovation activities'. *Journal of Business and Industrial Marketing*, **28**, 444–54.

Chien, S. Y. and Tsai, C. H. (2012). 'Dynamic capability, knowledge, learning, and firm performance'. *Journal of Organizational Change Management*, **25**, 434–44.

Coff, R. W. (1999). 'When competitive advantage doesn't lead to performance: the resource-based view and stakeholder bargaining power'. *Organization Science*, **10**, 119–33.

Collis, D. J. (1994). 'How valuable are organizational capabilities?'. *Strategic Management Journal*, **24**, 143–52.

Crook, T. R., Ketchen, D. J., Combs, J. G. and Todd, S. Y. (2008). 'Strategic resources and performance: a meta-analysis'. *Strategic Management Journal*, **29**, 1141–54.

Cuervo-Cazurra, A. (2006). 'Who cares about corruption?'. *Journal of International Business Studies*, **37**, 807–22.

Cuervo-Cazurra, A. (2012). 'Extending theory by analyzing developing country multinational companies: solving the Goldilocks debate'. *Global Strategy Journal*, **2**, 153–67.

Cui, Y. and Jiao, H. (2011). 'Dynamic capabilities, strategic stakeholder alliances and sustainable competitive advantage: evidence from China'. *Corporate Governance*, **11**, 386–98.

Daily, C. M., Certo, S. T. and Dalton, D. R. (2005). 'Investment bankers and IPO pricing: does prospectus information matter?'. *Journal of Business Venturing*, **20**, 93–111.

Danneels, E. (2002). 'The dynamics of product innovation and firm competences'. *Strategic Management Journal*, **23**, 1095–121.

Danneels, E. (2008). 'Organizational antecedents of second-order competences'. *Strategic Management Journal*, **29**, 519–43.

Danneels, E. (2012). 'Second-order competences and Schumpeterian rents'. *Strategic Entrepreneurship Journal*, **6**, 42–58.

Danneels, E. (2015). 'Survey measures of first-and second order competences'. *Strategic Management Journal*. Available at http://onlinelibrary.wiley.com/doi/10.1002/smj.2428/full.

David, R. J. and Han, S. K. (2004). 'A systematic assessment of the empirical support for transaction cost economics'. *Strategic Management Journal*, **25**, 39–58.

Deeds, D. L., DeCarolis, D. M. and Coombs, J. (2000). 'Dynamic capabilities and new product development in high technology ventures: an empirical analysis of new biotechnology firms'. *Journal of Business Venturing*, **15**, 211–29.

Dess, G. G. and Beard, D. W. (1984). 'Dimensions of organizational task environments'. *Administrative Science Quarterly*, **29**, 52–73.

Di Giovanni, J. (2005). 'What drives capital flows? The case of cross-border M&a activity and financial deepening'. *Journal of International Economics*, **65**, 127–49.

Di Stefano, G., Peteraf, M. and Verona, G. (2010). 'Dynamic capabilities deconstructed: a bibliographic investigation into the origins, development, and future directions of the research domain'. *Industrial and Corporate Change*, **19**, 1187–204.

Dixon, S. E., Meyer, K. E. and Day, M. (2010). 'Stages of organizational transformation in transition economies: a dynamic capabilities approach'. *Journal of Management Studies*, **47**, 416–36.

Drnevich, P. L. and Kriauciunas, A. P. (2011). 'Clarifying the conditions and limits of the contributions of ordinary and dynamic capabilities to relative firm performance'. *Strategic Management Journal*, **32**, 254–79.

Dunning, J. H. (1980). 'Towards an eclectic theory of international production: some empirical tests'. *Journal of International Business Studies*, **11**, 9–31.

Dunning, J. H. (2000). 'The eclectic paradigm as an envelope for economic and business theories of MNE activity'. *International Business Review*, **9**, 163–90.

Dunning, J. H. and Lundan, S. M. (2010). 'The institutional origins of dynamic capabilities in multinational enterprises'. *Industrial and Corporate Change*, **19**, 1225–46.

Duval, S. and Tweedie, R. (2000a). 'Trim and fill: a simple funnel-plot–based method of testing and adjusting for publication bias in meta-analysis'. *Biometrics*, **56**, 455–63.

Duval, S. and Tweedie, R. (2000b). 'A nonparametric "trim and fill" method of accounting for publication bias in meta-analysis'. *Journal of the American Statistical Association*, **95**, 89–98.

Dyer, J. H. and Hatch, N. W. (2006). Relation-specific capabilities and barriers to knowledge transfers: creating advantage through network relationships'. *Strategic Management Journal*, **27**, 701–19. '

Easterby-Smith, M., Lyles, M. and Peteraf, M. (2009). 'The practice of dynamic capabilities'. *British Journal of Management*, **20**, 1–8. Special Issue,

Eggers, J. P. (2012). 'All experience is not created equal: learning, adapting, and focusing in product portfolio management'. *Strategic Management Journal*, **33**, 315–35.

Eisenhardt, K. M. and Martin, J. (2000). 'Dynamic capabilities: what are they?'. *Strategic Management Journal*, **21**, 1105–21.

Eisenhardt, K. M., Furr, N. R. and Bingham, C. B. (2010). 'CROSSROADS-Microfoundations of performance: Balancing efficiency and flexibility in dynamic environments'. *Organization Science*, **21**, 1263–73.

Eng, T. Y. and Spickett-Jones, J. G. (2009). 'An investigation of marketing capabilities and upgrading performance of manufacturers in mainland China and Hong Kong'. *Journal of World Business*, **44**, 463–75.

Ettlie, J. E. and Pavlou, P. A. (2006). 'Technology-based new product development partnerships'. *Decision Sciences*, **37**, 117–47.

Fang, E. E. and Zou, S. (2009). 'Antecedents and consequences of marketing dynamic capabilities in international joint ventures'. *Journal of International Business Studies*, **40**, 742–61.

Fernández-Mesa, A., Alegre-Vidal, J., Chiva-Gómez, R. and Gutiérrez-Gracia, A. (2013). 'Design management capability and product innovation in SMEs'. *Management Decision*, **51**, 547–65.

Filatotchev, I. and Piesse, J. (2009). 'R&D, internationalization and growth of newly listed firms: European evidence'. *Journal of International Business Studies*, **40**, 1260–76.

Franco, A. M., Sarkar, M. B., Agarwal, R. and Echambadi, R. (2009). 'Swift and smart: the moderating effects of technological capabilities on the market pioneering-firm survival relationship'. *Management Science*, **55**, 1842–60.

García-Morales, V. J., Jiménez-Barrionuevo, M. M. and Gutiérrez-Gutiérrez, L. (2012). 'Transformational leadership influence on organizational performance through organizational learning and innovation'. *Journal of Business Research*, **65**, 1040–50.

Geyskens, I., Krishnan, R., Steenkamp, J. B. E. and Cunha, P. V. (2009). 'A review and evaluation of meta-analysis practices in management research'. *Journal of Management*, **35**, 393–419.

Godfrey, P. C. and Hill, C. W. (1995). 'The problem of unobservables in strategic management research'. *Strategic Management Journal*, **16**, 519–33.

Guillen, M. F. (2000). 'Business groups in emerging economies: a resource-based view'. *Academy of Management Journal*, **43**, 362–80.

Harrison, J. S., Banks, G. C., Pollack, J. M., O'boyle, E. H. and Short, J. (2014). 'Publication bias in strategic management research'. *Journal of Management*, doi: 10.1177/0149206314535438.

He, X. and Cui, L. (2012). 'Can strong home country institutions foster the internationalization of MNEs?'. *Multinational Business Review*, **20**, 352–75.

Hedges, L. V.,. and Olkin, I. (1985). *Statistical Methods for Meta-Analysis*. Orlando, FL: Academic Press.

Heimeriks, K. H., Schijven, M. and Gates, S. (2012). 'Manifestations of higher-order routines: the underlying mechanisms of deliberate learning in the context of post-acquisition integration'. *Academy of Management Journal*, **55**, 703–26.

Helfat, C. E. and Martin, J. A. (2015). 'Dynamic managerial capabilities: review and assessment of managerial impact on strategic change'. *Journal of Management*, **41**, 1281–312.

Helfat, C. E. and Peteraf, M. A. (2009). 'Understanding dynamic capabilities: progress along a developmental path'. *Strategic Organization*, **7**, 91–10.

Helfat, C. E. and Peteraf, M. A. (2014). 'Managerial cognitive capabilities and the microfoundations of dynamic capabilities'. *Strategic Management Journal*, **36**, 831–50.

Helfat, C. E. and Winter, S. G. (2011). 'Untangling dynamic and operational capabilities: strategy for the (n)ever-changing world'. *Strategic Management Journal*, **32**, 1243–50.

Helfat, C. E., Finkelstein, S., Mitchell, W., Peteraf, M. A., Singh, H., Teece, D. J. and Winter, S. G. (2007). *Dynamic Capabilities: Understanding Strategic Change in Organizations*. London: Blackwell.

Hoskisson, R. E., Eden, L., Lau, C. M. and Wright, M. (2000). 'Strategy in emerging economies'. *Academy of Management Journal*, **43**, 249–67.

Hsu, I. and Sabherwal, R. (2012). 'Relationship between intellectual capital and knowledge management: an empirical investigation'. *Decision Sciences*, **43**, 489–524.

Hsu, L. C. and Wang, C. H. (2012). 'Clarifying the effect of intellectual capital on performance: the mediating role of dynamic capability'. *British Journal of Management*, **23**, 179–205.

Huang, K. F., Wu, L. Y., Dyerson, R. and Chen, C. F. (2012). 'How does a technological firm develop its competitive advantage? A dynamic capability perspective'. *IEEE Transactions on Engineering Management*, **59**, 644–53.

Huh, S., Yook, K. H. and Kim, I. W. (2008). 'Relationship between organizational capabilities and performance of target costing: an empirical study of Japanese companies'. *Journal of International Business Research*, **7**, 91–107.

Hult, G. T. M. and Ketchen, D. J. (2001). 'Does market orientation matter? A test of the relationship between positional advantage and performance'. *Strategic Management Journal*, **22**, 899–906.

Hung, R. Y. Y., Yang, B., Lien, B. Y. H., McLean, G. N. and Kuo, Y. M. (2010). 'Dynamic capability: impact of process alignment and organizational learning culture on performance'. *Journal of World Business*, **45**, 285–94.

Hunter, J. E. and Schmidt, F. L. (1990). 'Dichotomization of continuous variables: the implications for meta-analysis'. *Journal of Applied Psychology*, **75**, 334–49.

Hunter, J. E. and Schmidt, F. L. (2000). 'Fixed effects vs. random effects meta-analysis models: implications for cumulative research knowledge'. *International Journal of Selection and Assessment*, **8**, 275–92.

Hunter, J. E. and Schmidt, F. L. (2004). *Methods of Meta-Analysis: Correcting Error and Bias in Research Findings*, 2nd edition. Newbury Park, CA: Sage Publications.

Jacobides, M. G. and Winter, S. G. (2005). 'The co-evolution of capabilities and transaction costs: explaining the institutional structure of production'. *Strategic Management Journal*, **26**, 395–413.

Jantunen, A., Puumalainen, K., Saarenketo, S. and Kyläheiko, K. (2005). 'Entrepreneurial orientation, dynamic capabilities and international performance'. *Journal of International Entrepreneurship*, **3**, 223–43.

Jarratt, D. (2004). 'Conceptualizing a relationship management capability'. *Marketing Theory*, **4**, 287–309.

Jiang, R. J., Tao, Q. T. and Santoro, M. D. (2010). 'Alliance portfolio diversity and firm performance'. *Strategic Management Journal*, **31**, 1136–44.

Kale, P. and Singh, H. (2007). 'Building firm capabilities through learning: the role of the alliance learning process in alliance capability and firm-level alliance success'. *Strategic Management Journal*, **28**, 981–1000.

Karim, S. (2009). 'Business unit reorganization and innovation in new product markets'. *Management Science*, **55**, 1237–54.

Khalid, S. and Larimo, J. (2012). 'Firm specific advantage in developed markets dynamic capabilities perspective'. *Management International Review*, **52**, 233–50.

Khanna, T. and Palepu, K. (1997). 'Why focused strategies may be wrong for emerging markets'. *Harvard Business Review*, **75**, 41–54.

Khatri, N., Ohja, A. K. and Gupta, V. (2014). *HR and IT Capabilities and Complementarities in U.S. health care organizations*. Working paper.

Kim, H., Kim, H. and Hoskisson, R. E. (2010). 'Does market-oriented institutional change in an emerging economy make business-group-affiliated multinationals perform better? An institution-based view'. *Journal of International Business Studies*, **41**, 1141–60.

Kim, M., Suresh, N. C. and Kocabasoglu-Hillmer, C. (2013). 'The impact of manufacturing flexibility and technological dimensions of manufacturing strategy on improving supply chain responsiveness: business environment perspective'. *International Journal of Production Research*, **51**, 5597–611.

Kirca, A., Hult, G. T., Roth, K., Cavusgil, S. T., Perry, M. Z., Akdeniz, M. B., Deligonul, S. Z., Mena, J. A., Pollitte, W. A., Hoppner, J. J., Miller, M. C. and White, R. C. (2011). 'Firm-specific assets, multinationality, and financial performance: a meta-analytic and theoretical integration'. *Academy of Management Journal*, **54**, 47–72.

Kirca, A. H., Roth, K., Hult, G. T. M. and Cavusgil, S. T. (2012). 'The role of context in the multinationality-performance relationship: a meta-analytic review'. *Global Strategy Journal*, **2**, 108–21.

Kor, Y. Y. and Mahoney, J. T. (2005). 'How dynamics, management, and governance of resource deployments influence firm-level performance'. *Strategic Management Journal*, **26**, 489–96.

Kotha, R., Zheng, Y. and George, G. (2011). 'Entry into new niches: the effects of firm age and the expansion of technological capabilities on innovative output and impact'. *Strategic Management Journal*, **32**, 1011–24.

Krasnikov, A. and Jayachandran, S. (2008). 'The relative impact of marketing, research-and-development, and operations capabilities on firm performance'. *Journal of Marketing*, **72**, 1–11.

Lavie, D. (2006). 'Capability reconfiguration: an analysis of incumbent responses to technological change'. *Academy of Management Review*, **31**, 153–74.

Lawler, J. J., Chen, S. J., Wu, P. C., Bae, J. and Bai, B. (2011). 'High-performance work systems in foreign subsidiaries of American multinationals: an institutional model'. *Journal of International Business Studies*, **42**, 202–20.

Lee, C., Venkatraman, N., Tanriverdi, H. and Iyer, B. (2010). Complementarity-based hypercompetition in the software industry: theory and empirical test, 1990–2002. *Strategic Management Journal*, **31**, 1431–56.*

Lee, J. and Slater, J. (2007). 'Dynamic capabilities, entrepreneurial rent-seeking and the investment development path: the case of Samsung'. *Journal of International Management*, **13**, 241–57.

Lee, R. P., Naylor, G. and Chen, Q. (2011). 'Linking customer resources to firm success: the role of marketing program implementation'. *Journal of Business Research*, **64**, 394–400.

Li, D. Y. and Liu, J. (2014). 'Dynamic capabilities, environmental dynamism, and competitive advantage: evidence from China'. *Journal of Business Research*, **67**, 2793–99.

Liu, H. Y. and Hsu, C. W. (2011). 'Antecedents and consequences of corporate diversification: a dynamic capabilities perspective'. *Management Decision*, **49**, 1510–34.

Lockett, A., Thompson, S. and Morgenstern, U. (2009). 'The development of the resource-based view of the firm: a critical appraisal'. *International Journal of Management Reviews*, **11**, 9–28.

Lu, Y. and Ramamurthy, K. (2010). 'Proactive or reactive IT leaders and quest; a test of two competing hypotheses of IT innovation and environment alignment'. *European Journal of Information Systems*, **19**, 601–18.

Luo, Y. (2003a). 'Industrial dynamics and managerial networking in an emerging market: the case of China'. *Strategic Management Journal*, **24**, 1315–27.

Luo, Y. (2003b). 'Market-seeking MNEs in an emerging market: how parent–subsidiary links shape overseas success'. *Journal of International Business Studies*, **34**, 290–309.

Luo, Y. and Tung, R. L. (2007). 'International expansion of emerging market enterprises: a springboard perspective'. *Journal of International Business Studies*, **38**, 481–98.

Makadok, R. (2011). 'Invited editorial: the four theories of profit and their joint effects'. *Journal of Management*, **37**, 1316–34.

Makkonen, H., Pohjola, M., Olkkonen, R. and Koponen, A. (2014). 'Dynamic capabilities and firm performance in a financial crisis'. *Journal of Business Research*, **67**, 2707–19.

Malik, O. R. (2008). 'Adapting to market liberalization: the role of dynamic capabilities, initial resource conditions, and strategic path choices in determining evolutionary fitness of Less Developed Country (LDC) firms'. *Journal of International Management*, **14**, 217–31.

Malik, O. R. and Kotabe, M. (2009). 'Dynamic capabilities, government policies, and performance in firms from emerging economies: evidence from India and Pakistan'. *Journal of Management Studies*, **46**, 421–50.

Marcus, A. A. and Anderson, M. H. (2006). 'A general dynamic capability: does it propagate business and social competencies in the retail food industry?'. *Journal of Management Studies*, **43**, 19–46.

Maritan, C. A. (2001). 'Capital investment as investing in organizational capabilities: an empirically grounded process model'. *Academy of Management Journal*, **44**, 513–31.

Marquis, C. and Raynard, M. (2015). 'Institutional strategies in emerging markets'. *Academy of Management Annals*, **9**, 291–335.

Mathews, J. A. (2006). Dragon multinationals: new players in 21st century globalization. *Asia Pacific Journal of Management*, **23**, 5–27.

McCarthy, I. P., Lawrence, T. B., Wixted, B. and Gordon, B. R. (2010). 'A multidimensional conceptualization of environmental velocity'. *Academy of Management Review*, **35**, 604–26.

McGrath, R. G. (2013). 'Transient advantage'. *Harvard Business Review*, **91**, 62–70.

Menguc, B. and Auh, S. (2006). 'Creating a firm-level dynamic capability through capitalizing on market orientation and innovativeness'. *Journal of the Academy of Marketing Science*, **34**, 63–73.

Moliterno, T. P. and Wiersema, M. F. (2007). 'Firm performance, rent appropriation, and the strategic resource divestment capability'. *Strategic Management Journal*, **28**, 1065–87.

Morgan, N. A., Vorhies, D. W. and Mason, C. H. (2009). 'Market orientation, marketing capabilities, and firm performance'. *Strategic Management Journal*, **30**, 909–20.

Morgan, N. A., Katsikeas, C. S. and Vorhies, D. W. (2012). 'Export marketing strategy implementation, export marketing capabilities, and export venture performance'. *Journal of the Academy of Marketing Science*, **40**, 271–89.

Morrison, A., Pietrobelli, C. and Rabellotti, R. (2008). 'Global value chains and technological capabilities: a framework to study learning and innovation in developing countries'. *Oxford Development Studies*, **36**, 39–58.

Nair, A., Rustambekov, E., McShane, M. and Fainshmidt, S. (2014). 'Enterprise risk management as a dynamic capability: a test of its effectiveness during a crisis'. *Managerial and Decision Economics*, **35**, 555–66.

Nelson, R. R. and Winter, S. G. (1982). 'The Schumpeterian tradeoff revisited'. *American Economic Review*, **72**, 114–32.

Newbert, S. L. (2007). 'Empirical research on the resource-based view of the firm: an assessment and suggestions for future research'. *Strategic Management Journal*, **28**, 121–46.

Orwin, R. G. (1983). 'A fail-safe N for effect size in meta-analysis'. *Journal of Educational Statistics*, **8**, 157–9.

O'Cass, A. and Sok, P. (2012). 'Examining the role of within functional area resource–capability complementarity in achieving customer and product-based performance outcomes'. *Journal of Strategic Marketing*, **20**, 345–63.

Pablo, A., Reay, T., Dewald, J. R. and Casebeer, A. L. (2007). 'Identifying, enabling and managing dynamic capabilities in the public sector'. *Journal of Management Studies*, **44**, 687–708.

Palich, L. E., Cardinal, L. B. and Miller, C. C. (2000). 'Curvilinearity in the diversification-performance linkage: an examination of over three decades of research'. *Strategic Management Journal*, **21**, 155–74.

Pavlou, P. A. and El Sawy, O. A. (2010). 'The "third hand": IT-enabled competitive advantage in turbulence through improvisational capabilities'. *Information Systems Research*, **21**, 443–71.

Peng, M. W. (2003). 'Institutional transitions and strategic choices'. *Academy of Management Review*, **28**, 275–96.

Peng, M. W., Zhang, S. and Li, X. (2007). 'CEO duality and firm performance during China's institutional transitions'. *Management and Organization Review*, **3**, 205–25.

Peng, M. W., Sun, S. L., Pinkham, B. and Chen, H. (2009). 'The institution-based view as a third leg for a strategy tripod'. *Academy of Management Perspectives*, **23**, 63–81.

Peteraf, M. A. and Barney, J. B. (2003). 'Unraveling the resource-based tangle'. *Managerial and Decision Economics*, **24**, 309–23.

Peteraf, M., Di Stefano, G. and Verona, G. (2013). 'The elephant in the room of dynamic capabilities: bringing two diverging conversations together'. *Strategic Management Journal*, **34**, 1389–410.

Protogerou, A., Caloghirou, Y. and Lioukas, S. (2012). 'Dynamic capabilities and their indirect impact on firm performance'. *Industrial and Corporate Change*, **21**, 615–47.

Protogerou, A., Caloghirou, Y. D. and Karagouni, G. (2014). 'The relevance of the "dynamic capabilities" perspective in low-tech sectors'. In Hirsch-Kreinsen, H. and Schwinge, I. (Eds), *Knowledge-Intensive Entrepreneurship in Low-Tech Industries*. Cheltenham: Edward Elgar, 138–65.

Rahmandad, H. (2012). 'Impact of growth opportunities and competition on firm-level capability development trade-offs'. *Organization Science*, **23**, 138–54.

Ramamurti, R. (2012). 'What is really different about emerging market multinationals?'. *Global Strategy Journal*, **2**, 41–7. '

Ray, G., Barney, J. B. and Muhanna, W. A. (2004). 'Capabilities, business processes, and competitive advantage: choosing the dependent variable in empirical tests of the resource-based view'. *Strategic Management Journal*, **25**, 23–37.

Reed, R. and DeFillippi, R. J. (1990). 'Causal ambiguity, barriers to imitation, and sustainable competitive advantage'. *Academy of Management Review*, **15**, 88–102.

Rice, J., Liao, T. S., Galvin, P. and Martin, N. (2015). 'A configuration-based approach to integrating dynamic capabilities and market transformation in small and medium-sized enterprises to achieve firm performance'. *International Small Business Journal*, **33**, 231–53.

Rindova, V. P. and Kotha, S. (2001). 'Continuous "morphing": competing through dynamic capabilities, form, and function'. *Academy of Management Journal*, **44**, 1263–80.

Roberts, N. and Grover, V. (2012). 'Investigating firm's customer agility and firm performance: the importance of aligning sense and respond capabilities'. *Journal of Business Research*, **65**, 579–85.

Romme, A. G. L., Zollo, M. and Berends, P. (2010). 'Dynamic capabilities, deliberate learning and environmental dynamism: a simulation model'. *Industrial and Corporate Change*, **19**, 1271–99.

Rosenberg, M. S. (2005). 'The file-drawer problem revisited: a general weighted method for calculating fail-safe numbers in meta-analysis'. *Evolution*, **59**, 464–8.

Rothaermel, F. T. and Hess, A. M. (2007). 'Building dynamic capabilities: innovation driven by individual-, firm-, and network-level effects'. *Organization Science*, **18**, 898–921.

Salge, O. and Vera, A. (2013). 'Small steps that matter: incremental learning, slack resources and organizational performance'. *British Journal of Management*, **24**, 156–73.

Salvato, C. and Rerup, C. (2011). 'Beyond collective entities: multilevel research on organizational routines and capabilities'. *Journal of Management*, **37**, 468–90.

Santos-Vijande, M. L., López-Sánchez, J. Á. and Trespalacios, J. A. (2011). 'How organizational learning affects a firm's flexibility, competitive strategy, and performance'. *Journal of Business Research*, **65**, 1079–89.

Santos-Vijande, M. L., del Río-Lanza, A. B., Suárez-Álvarez, L. and Díaz-Martín, A. M. (2013). 'The brand management system and service firm competitiveness'. *Journal of Business Research*, **66**, 148–57.

Sarkar, M. B., Echambadi, R. A. J. and Harrison, J. S. (2001). 'Alliance entrepreneurship and firm market performance'. *Strategic Management Journal*, **22**, 701–11.

Schilke, O. (2014a). 'On the contingent value of dynamic capabilities for competitive advantage: the nonlinear moderating effect of environmental dynamism'. *Strategic Management Journal*, **35**, 179–203.

Schilke, O. (2014b). 'Second-order dynamic capabilities: how do they matter?'. *Academy of Management Perspectives*, **28**, 368–80.

Schmidt, F. L., Oh, I. and Hayes, T. L. (2009). 'Fixed-versus random effects models in meta-analysis: model properties and an empirical comparison of differences in results'. *British Journal of Mathematical and Statistical Psychology*, **62**, 97–128.

Schreyögg, G. and Kliesch-Eberl, M. (2007). 'How dynamic can organizational capabilities be? towards a dual-process model of capability dynamization'. *Strategic Management Journal*, **28**, 913–33.
Shamsie, J., Martin, X. and Miller, D. (2009). 'In with the old, in with the new: capabilities, strategies, and performance among the Hollywood studios'. *Strategic Management Journal*, **30**, 1440–52.
Shinkle, G. A. and Kriauciunas, A. P. (2012). 'The impact of current and founding institutions on strength of competitive aspirations in transition economies'. *Strategic Management Journal*, **33**, 448–58.
Simerly, R. L. and Li, M. (2000). 'Environmental dynamism, capital structure and performance: a theoretical integration and an empirical test'. *Strategic Management Journal*, **21**, 31–50.
Singh, D., Oberoi, J. S. and Ahuja, I. S. (2013). 'An empirical investigation of dynamic capabilities in managing strategic flexibility in manufacturing organizations'. *Management Decision*, **51**, 1442–61.
Slater, S. F. and Narver, J. C. (1994). 'Does Competitive environment moderate the market orientation-performance relationship?'. *Journal of Marketing*, **58**, 46–55.
Song, M., Droge, C., Hanvanich, S. and Calantone, R. (2005). 'Marketing and technology resource complementarity: an analysis of their interaction effect in two environmental contexts'. *Strategic Management Journal*, **26**, 259–59.
Stadler, C., Helfat, C. E. and Verona, G. (2013). 'The impact of dynamic capabilities on resource access and development'. *Organization Science*, **24**, 1782–804.
Teece, D. J. (2007). 'Explicating dynamic capabilities: the nature and microfoundations of (sustainable) enterprise performance'. *Strategic Management Journal*, **28**, 1319–50.
Teece, D. (2014). 'The foundations of enterprise performance: dynamic and ordinary capabilities in an (economic) theory of firms'. *Academy of Management Perspectives*, **28**, 328–52.
Teece, D., Pisano, G. and Shuen, A. (1997). 'Dynamic capabilities and strategic management'. *Strategic Management Journal*, **18**, 509–33.
Thornhill, S. (2006). 'Knowledge, innovation and firm performance in high-and low-technology regimes'. *Journal of Business Venturing*, **21**, 687–703.
Tripsas, M. (1997). 'Unraveling the process of creative destruction: complementary assets and incumbent survival in the typesetter industry'. *Strategic Management Journal*, **18**, 119–42.
Uhlaner, L. M., van Stel, A., Duplat, V. and Zhou, H. (2013). 'Disentangling the effects of organizational capabilities, innovation and firm size on SME sales growth'. *Small Business Economics*, **41**, 581–607.
Uhlenbruck, K. (2004). 'Developing acquired foreign subsidiaries: the experience of MNEs in transition economies'. *Journal of International Business Studies*, **35**, 109–23.
Uhlenbruck, K., Meyer, K. E. and Hitt, M. A. (2003). 'Organizational transformation in transition economies: resource-based and organizational learning perspectives'. *Journal of Management Studies*, **40**, 257–82.
Venkatraman, N. and Ramanujam, V. (1986). 'Measurement of business performance in strategy research: a comparison of approaches'. *Academy of Management Review*, **11**, 801–14.
Vergne, J. P. and Durand, R. (2011). 'The path of most persistence: an evolutionary perspective on path dependence and dynamic capabilities'. *Organization Studies*, **32**, 365–82.
Verona, G. (1999). 'A resource-based view of product development'. *Academy of Management Review*, **24**, 132–42.
Vickery, S. K., Koufteros, X. and Droge, C. (2013). 'Does product platform strategy mediate the effects of supply chain integration on performance? A dynamic capabilities perspective'. *IEEE Transactions on Engineering Management*, **60**, 750–62.
Viechtbauer, W. (2010). 'Conducting meta-analyses in R with the metafor package'. *Journal of Statistical Software*, **36**, 1–48.
Viswesvaran, C. and Ones, D. S. (1995). 'Theory testing: combining psychometric meta-analysis and structural equations modeling'. *Personnel Psychology*, **48**, 865–85.
Vogel, R. and Güttel, W. H. (2013). 'The dynamic capability view in strategic management: a bibliometric review'. *International Journal of Management Reviews*, **15**, 426–46.
Wang, C. K. and Ang, B. L. (2004). 'Determinants of venture performance in Singapore'. *Journal of Small Business Management*, **42**, 347–63.
Wang, C. L. and Ahmed, P. K. (2007). 'Dynamic capabilities: a review and research agenda'. *International Journal of Management Reviews*, **9**, 31–51.
Wang, C. L., Senaratne, C. and Rafiq, M. (2014). 'Success traps, dynamic capabilities and firm performance'. *British Journal of Management*, **26**, 26–44.
Webb, B. R. and Schlemmer, F. (2008). 'Predicting web services performance from internet performance: an empirical study of resources and capabilities in e-business SMEs'. *Journal of Knowledge Management*, **12**, 137–55.

Weerawardena, J., Mort, G. S., Liesch, P. W. and Knight, G. (2007). 'Conceptualizing accelerated internationalization in the born global firm: a dynamic capabilities perspective'. *Journal of World Business*, **42**, 294–306.

Wei, H. L. and Wang, E. T. (2010). 'The strategic value of supply chain visibility: increasing the ability to reconfigure'. *European Journal of Information Systems*, **19**, 238–49.

Wernerfelt, B. (1984). 'A resource-based view of the firm'. *Strategic Management Journal*, **5**, 171–80.

Whitener, E. M. (1990). 'Confusion of confidence intervals and credibility intervals in meta-analysis'. *Journal of Applied Psychology*, **75**, 315–21.

Wilden, R. and Gudergan, S. P. (2014). 'The impact of dynamic capabilities on operational marketing and technological capabilities: investigating the role of environmental turbulence'. *Journal of the Academy of Marketing Science*, **43**, 181–99.

Wilden, R., Gudergan, S. P., Nielsen, B. B. and Lings, I. (2013). 'Dynamic capabilities and performance: strategy, structure and environment'. *Long Range Planning*, **46**, 72–96.

Williams, L. J., Hartman, N. and Cavazotte, F. (2010). 'Method variance and marker variables: a review and comprehensive CFA marker technique'. *Organizational Research Methods*, **13**, 477–514.

Winter, S. G. (2003). 'Understanding dynamic capabilities'. *Strategic Management Journal*, **24**, 991–5.

Winter, S. G. (2012). 'Capabilities: their origins and ancestry'. *Journal of Management Studies*, **49**, 1402–6.

Wu, L. Y. (2006). 'Resources, dynamic capabilities and performance in a dynamic environment: perceptions in Taiwanese IT enterprises'. *Original Research Article Information and Management*, **43**, 447–54.

Wu, L. Y. (2007). 'Entrepreneurial resources, dynamic capabilities and start-up performance of Taiwan's high-tech firms'. *Journal of Business Research*, **60**, 549–55.

Wu, L. Y. (2010). 'Applicability of the resource-based and dynamic-capability views under environmental volatility'. *Journal of Business Research*, **63**, 27–31.

Wu, S. H., Lin, L. Y. and Hsu, M. Y. (2007). 'Intellectual capital, dynamic capabilities and innovative performance of organizations'. *International Journal of Technology Management*, **39**, 279–96.

Yalcinkaya, G., Calantone, R. and Griffith, D. (2007). 'An examination of exploration and exploitation capabilities: implications for product innovation and market performance'. *Journal of International Marketing*, **15**, 63–93.

Yiu, D. W. and Lau, C. M. (2008). 'Corporate entrepreneurship as resource capital configuration in emerging market firms'. *Entrepreneurship Theory and Practice*, **32**, 37–57.

Yiu, D. W., Lau, C. and Bruton, G. D. (2007). 'International venturing by emerging economy firms: the effects of firm capabilities, home country networks, and corporate entrepreneurship'. *Journal of International Business Studies*, **38**, 519–40.

Zahra, S. A. and George, G. (2002). 'Absorptive capacity: a review, reconceptualization, and extension'. *Academy of Management Review*, **27**, 185–203.

Zahra, S. A. and Neubaum, D. O. (1998). 'Environmental adversity and the entrepreneurial activities of new ventures'. *Journal of Developmental Entrepreneurship*, **3**, 123–40.

Zahra, S. A., Sapienza, H. J. and Davidsson, P. (2006). 'Entrepreneurship and dynamic capabilities: a review, model and research agenda'. *Journal of Management Studies*, **43**, 917–55.

Zhan, W. and Chen, R. R. (2013). 'Dynamic capability and IJV performance: the effect of exploitation and exploration capabilities'. *Asia Pacific Journal of Management*, **30**, 601–32.

Zhan, W. and Luo, Y. (2008). 'Performance implications of capability exploitation and upgrading in international joint ventures'. *Management International Review*, **48**, 227–53.

Zollo, M. and Winter, S. G. (2002). 'Deliberate learning and the evolution of dynamic capabilities'. *Organization Science*, **13**, 339–51.

Zott, C. (2003). 'Dynamic capabilities and the emergence of intraindustry differential firm performance: insights from a simulation study'. *Strategic Management Journal*, **24**, 97–125.